What a Verifiable Reward Can and Cannot Credit in Self-Extending Agent Memory: A Non-Identifiability Result for False-Lesson Harm

James Hugglestone Samuel Jacob Chacko Xiuwen Liu (adv.)
Florida State University — *preprint*

Abstract

A self-extending memory lets an LLM agent reuse what it has learned, but consolidating its own experience can *degrade* performance when the agent encodes causally-incorrect lessons—**false-lesson propagation**. We study this in a *verifiable-reward* domain, Capture-The-Flag, and ask: can the verifiable flag alone—no gold trajectory, no human label, no LLM judge—tell a genuinely useful lesson from a causally-false (*poison*) one the agent *executes and recovers from*? We map the boundary on a controlled testbed whose lesson causality is known by construction. The agent's stochastic retrieval is a *free randomized experiment*—each lesson's inclusion is a draw with a logged propensity—so inverse-propensity weighting over its own logs recovers, at no extra inference, the large value of *helpful* lessons that outcome credit conflates with poison. Our central result is *negative*: the per-item harm of a benign-false lesson is *not identifiable from outcomes*. Neither the cheap estimate nor an assumption-light leave-one-out gold standard can rank it (gold-standard mean $+0.045$, 95 % CI $[-0.006, +0.098]$), and the mildly-negative observational sign is confounded, not the causal estimand. We give a scoped argument why: under a measured base-rate floor, executing a benign-wrong recipe is, for the binary outcome, nearly equivalent to executing none. A positive control delimits what *can* be caught: actively-destructive lessons are detected, a confidently-wrong answer is untested (the agent almost never ran it), and a verifier recovers recipe validity by construction. "Non-identifiable" means *not per-item-rankable*, not "harmless": a small sub-floor harm persists when poison runs, and the null is scoped to one crypto-decode microbenchmark (a second, SQL, one breaks it; a third degenerates). The contribution is this boundary, stated honestly, plus a released, contamination-free, ground-truth-labeled instrument.

1 Introduction

Agentic LLM systems increasingly carry a *memory* that grows with use: skills, workflows, or lessons distilled from past trajectories and retrieved on new tasks [22, 28, 31, 34, 41]. The promise is compounding competence; the peril is compounding *error*. Recent work shows that consolidating even *flawless* solutions into reusable abstractions can *lower* an agent's success rate, because the abstraction step itself introduces incorrect causal lessons [33, 39]. The pessimistic reading is that self-consolidation should be avoided.

We argue the conclusion is domain-dependent. In open-ended domains the danger is acute precisely because success is *unverifiable*: a wrong lesson that an LLM judge accepts is silently absorbed. CTF challenges are different. A flag is a high-entropy string that the environment either accepts or rejects; success is *verifiable* and cheap to check. This removes the silent-corruption failure pole and leaves a sharper, measurable question: when a self-extending memory encodes a false lesson, can the verifiable signal alone—no gold trajectory, no human label, no LLM judge—*detect and suppress* it? The question is operational: every hygiene lever a deployed memory has—write-back gating, pruning, demotion—acts on *individual* lessons, so it needs a *per-item* credit signal; our central result is that the verifiable outcome cannot supply one for a benign-false lesson the agent recovers from.

We name the phenomenon **false-lesson propagation** and decompose it into three modes (§3): *birth defects* (a lesson trusted from a single success before its true reliability is known), *latent poison* (a lesson with a false causal premise that still wins, but via a longer route), and *recovery masking* (a lesson present during a success that did not cause it). Each mode is matched to an oracle-free mechanism in a system we build around a typed-node knowledge graph and a *contextual bandit* over its lessons (§4). Our contributions are:

1. A **boundary**: action-grounded credit handles a false lesson that is present but *unused*, and a promotion gate handles a *premature* one, but the verifiable signal *cannot* handle an *executed* false lesson the agent recovers from—a verifiable success credits whichever lesson was run. We reproduce this on real logs: an outcome-credit bandit values executed poison and true lessons indistinguishably.
2. A **causal-credit construction, and a precise account of its reach.** A verifiable-reward memory supplies, for free, the two ingredients causal identification needs: lesson inclusion is a stochastic draw with a *logged propensity*, and the reward is re-checkable without an oracle. We build the credit signal in three forms trading cost for precision—an active leave-one-out re-rollout (the assumption-light gold standard), an *observational* inverse-propensity estimate that reads the existing logs at *zero* extra inference,

and an online bandit that recomputes its values from its own logged exploration. This construction reliably identifies *helpful* lessons (large effects, larger at the high-capability rungs, non-monotonically across the full ladder)—the value outcome credit conflates with poison. But—our central, *negative* finding—it does *not* identify the per-item harm of a benign-false lesson the agent recovers from: neither the cheap estimate nor the gold standard can rank the per-item poison effect, and the observational sign is confounded. We give a scoped identification result for why (under a measured base-rate floor—exact at three capability rungs, above-floor at 3B, mildly violated at 32B—executing a benign-wrong recipe is, for the binary outcome, nearly equivalent to executing none), and a positive control that delimits the boundary—actively-destructive lessons are detected, a wrong-answer lesson the agent almost never followed is neither detected nor fairly tested, and a deterministic execution-verifier recovers *recipe validity*, a different and recoverable target.

3. An **analysis of where false lessons do and do not harm**, and a released instrument. The harm is not in immediate solve rate at deployed retrieval—a capable solver follows the true lesson when it is co-retrieved, so the credit signals sit at solve-parity (a small on-target harm remains; §9)—and, because the per-item poison value is non-identifiable from outcomes, no outcome-derived gate (the observational signal included) can be trusted to suppress it: the confounded label-level sign is not a working detector. We release a contamination-free micro-CTF instrument whose lesson causality is known by construction.[1]

Scope. The mechanism is demonstrated on a controlled micro-CTF instrument where lesson causality is known by construction; the boundary is *illustrated* by a 4B pilot on a single GPU and *established* on a lab-scale ladder of five capability rungs (Qwen2.5 3B–32B plus Llama-3.1-8B, §6). It is scoped to one crypto-decode microbenchmark: a second (SQL) author-built domain does not replicate the deployed-M null, and a third (struct-extraction) degenerates (§9). We report credit-signal quality, identification, and propagation—not absolute capability; the downstream cross-task solve-rate cost of propagation at scale, and whether the dynamics hold at the intended Qwen3.5-27B solver (which we did not run), are out of scope (§9).

2 Related Work

Self-evolving agent memory. Skill libraries and workflow memories let agents reuse experience [2, 20, 25, 28, 31, 32, 34, 41]. Zhang et al. [39] show consolidation can degrade performance and locate the fault in the abstraction step; Xiong et al. [33] trace the mechanism to similarity-retrieval copying that compounds error on hard cases. We take these as the null hypothesis our gated, credit-assigned design must beat. Bogdanov et al. [2] is the closest cyber-domain system (champion broadcast + failed-trajectory decomposition) but performs no false-lesson analysis; our released harness includes a re-implementation as a competitor, though we do not evaluate it here.

Credit assignment and the verifiable signal. Attributing outcome to the responsible step is hard [3, 16, 38, 40]. Per-item lesson credit is also the memory analogue of training-data valuation, where leave-one-out and Shapley-style per-example values are likewise small and noisy [9, 15]. Process-reward methods estimate step value from re-rollouts without human labels [29]; Yan et al. [36] make this rigorous for memory agents via local re-rollouts from a shared state—the oracle-free probe we adopt for our false-lesson-exposure metric. Verifiable rewards also power the mainstream RLVR training recipe [4, 17]; crucially, though, a verifiable reward is *not* a correct lesson: Yan et al. [35] show Qwen-family models can pass a checker via memorized shortcuts, and Helff et al. [10] document policies that satisfy a verifier while abandoning the intended method—our latent-poison and reward-hacking concerns. We therefore (i) condition reward on *step count*, not just outcome, and (ii) flag non-exploit captures explicitly.

Causal credit from logged randomization. Our solution rests on a classical identification idea applied to a new object. Inverse-propensity weighting recovers causal effects from logged data when treatment assignment has a known propensity [8, 19]; self-normalized weights and effective-sample-size diagnostics tame its variance [8]. Off-policy evaluation [19] and contextual-bandit OPE estimate the value of whole *policies*; we instead estimate the marginal value of each *individual memory item*, treating its stochastic retrieval as the randomized treatment. The closest prior is Yan et al. [36], which assigns memory credit by local re-rollouts from a shared state (LoGo-GRPO); our *active* probe is essentially their mechanism. Our contribution is the *observational* counterpart: in a verifiable-reward system the retrieval propensities are already logged, so the same causal credit is identifiable from the agent's own exploration at *zero* re-rollout cost—a cost–quality alternative to re-rolling, available precisely when reward is verifiable and inclusion is randomized with a known propensity. We frame this as the verifiable-reward setting supplying a *free randomized experiment*.

Retrieval, bandits, and off-policy evaluation. Growing the retrieved set to chase recall admits distractors that degrade long-context reasoning [13, 14]; we fix precision with a reranker [5] and diversity, not a larger pool (the reported runs use a deterministic lexical reranker; §6). Contextual bandits with a similarity prior personalize retrieval [6, 18]; logged propensities enable unbiased off-policy estimates [8, 19], which we use to sweep hyper-

[1] Code, the benchmark generator, and the committed experimental artifacts: `https://github.com/jawh3/solver-graph`.

parameters without spending the scarce GPU. Warm-starting a bandit from a prior helps only under low misalignment [1], motivating our cold-start treatment.

Evaluation discipline and contamination. A prior version of this project reported a +8.9pp improvement that was not statistically significant; we adopt paired, within-task comparisons with bootstrap confidence intervals and pre-run power analysis [7, 21]. Because public CTF write-ups and flags may be memorized, semantics-preserving challenge families separate *method learned* from *write-up memorized* [11]; our controlled corpus takes this to its limit with freshly-randomized flags. Grader-based difficulty is unreliable [12, 24], so we use ground-truth difficulty and a non-solver grader as a calibrated fallback. CTF agent benchmarks [23, 30, 37] and memory-poisoning analyses [26, 27, 42] situate the domain and the threat model.

3 The False-Lesson Propagation Problem

A *lesson* is a typed node (a tool, technique, or artifact observation) retrieved into the solver's context. A lesson is *false* when the causal premise it asserts is wrong for the challenge at hand. We distinguish three modes by *how the verifiable signal is fooled*:

Birth defect. A lesson distilled from a single solve is trusted before its reliability is known. Detection signal: per-node exposure count before it is allowed to influence runs at full weight.

Latent poison. A lesson with a false premise that still reaches the flag, but via a longer route than the correct method. Outcome-only credit cannot see it; *step count* in principle can, because the longer route shows up as more steps. Neither our pilot nor our lab-scale corpus instantiates this mode; we construct it in Appendix C, where the lever's target mode exists but the probe cannot resolve its step cost.

Recovery masking. A lesson sits in context during a successful run but is never used; the success came from a recovery pathway. Detection signal: whether the lesson's tool/technique appears in the agent's *executed actions*, not in its self-narration.

The claim we can support is *relative* and *bounded*: we measure and characterize false-lesson exposure *within the verifier-detectable subset*. Model-internal memorization shortcuts [35]—passing the checker without the lesson's stated mechanism—are out of scope and bounded by our contamination-free design (§6).

4 Method

The system retrieves a small set of lessons, solves with shell tools in a container, then updates a contextual bandit over the graph and optionally writes new lessons back. Figure 1 (and the released code) gives the full pipeline; here we describe the four false-lesson levers.

Coarse contextual bandit (birth-of-signal). Per `(context c, lesson v)` the bandit stores a value $Q[c][v]$ and an effective count $n[c][v]$. The context c is a *coarse* node—a CTF category or a k-means cluster over challenge embeddings—*not* a per-challenge identifier. A per-challenge context yields fewer than one update per cell within any realistic run budget and cannot learn; a sample-efficiency calculation (expected updates per cell $=\text{runs}\cdot M/(|c|\cdot\text{credited})$) gates the granularity. The sampling score keeps similarity as a decaying prior:

$$\text{score}(c, v) = \frac{n[c][v]\, Q[c][v] + \kappa\, \text{prior}(v)}{n[c][v] + \kappa}, \qquad (1)$$

where $\text{prior}(v)$ is the query–lesson cosine on the reward scale and κ controls how many observations override similarity. A cold lesson is sampled by its similarity, not an arbitrary uniform weight. We sample M lessons without replacement by sequential softmax at temperature τ and log the per-draw propensity (below).

Step-count reward (latent poison). With outcome $o \in \{0, 1\}$ and per-cell standardized steps z, efficiency is $e = -z$ and the run reward is

$$R = o\,(1 + \lambda e). \qquad (2)$$

A slow success earns $R < 1$, a fast success $R > 1$. A false lesson that wins via a longer route therefore accrues a lower Q than the correct method, and is down-weighted—without any label beyond the flag and the step count. Steps, not wall-clock, are used because wall-clock on a shared GPU is irreproducible.

Action-grounded credit (recovery masking). A sampled lesson v is *cited* iff its invocation signature (tool name, technique keywords, or produced artifact) appears in the agent's executed shell commands. Credit is assigned only to the cited subset $S = \text{sampled} \cap \text{cited}$:

$$Q[c][v] \mathrel{+}= \alpha\, g(d_v)\, \big((R - b) - Q[c][v]\big), \quad v \in S, \qquad (3)$$

with b the running mean reward in c and $g(d_v)$ a distance scaling. A lesson that merely sat in context during a success gains nothing—exactly denying the credit that recovery masking would otherwise capture. We do *not* use the model's self-report of which lessons it "used."

Gated write-back (birth defect). New lessons distilled from a trajectory enter as *provisional*: retrievable but weight-capped and flagged low-confidence in context. A provisional lesson is promoted to *trusted* only after it is cited in $\geq k$ successful runs across $\geq c$ distinct challenges. An N=1 lesson is therefore never trusted prematurely.

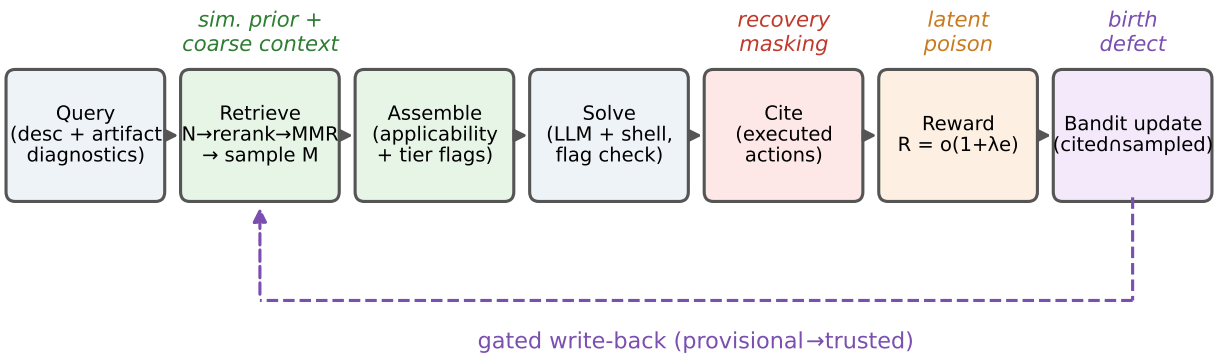


Figure 1: The solve-and-update loop. Each false-lesson lever (italics) attaches to one stage: a coarse contextual bandit with a similarity prior controls *which* lessons are sampled; action-grounded credit (cited∩sampled) attacks *recovery masking*; the step-count reward attacks *latent poison*; gated write-back attacks *birth defects*. Every sampling decision is logged with its propensity for off-policy evaluation.

Failures are decomposed too (tagged), and every new node carries a provenance edge and applicability conditions. Persistently-bad nodes are demoted rather than deleted, preserving an audit trail.

Logging and off-policy evaluation. Every run logs the candidate pool with its score components, the realized sampled set, the *per-draw* sampling propensities, and the sequential set log-probability. This makes the scarce GPU go further: an alternative policy π' (different M, τ, κ) is scored on already-collected runs by importance weighting, $w_i = P_{\pi'}/P_{\pi_{\log}}$, with inverse-propensity and doubly-robust estimators [8, 19]. We also implement a leave-one-out causal probe [36], in the process-reward-via-rollout tradition of Wang et al. [29]: re-rolling a solve with and without a suspect lesson, from a shared state, credits its *marginal* flag-reaching value—the oracle-free ground truth for false-lesson exposure.

5 From a Boundary to a Causal-Credit Construction

The levers of §4 run into a sharp boundary at *recovery masking* (§3). Action-grounded credit denies value to a false lesson that is never executed, and the step reward down-weights one that wins by a longer route; but a false lesson the agent *executes* and then recovers from receives the same verifiable-success credit as a genuinely useful one, because the flag credits whichever tool was run. The information needed to tell them apart—*would the run have succeeded without this lesson?*—is simply not in the scalar outcome of a single trajectory. §7 confirms this empirically: the outcome bandit's learned value cannot separate executed poison from true lessons. This section gives the construction that can.

Our key observation is that this counterfactual is, in principle, *estimable for free* in exactly this setting, for two reasons that coincide only in a verifiable-reward agentic-memory system. How far that estimability actually *reaches*—large helpful effects yes, the per-item harm of a benign-false lesson no—is the subject of §8 and the identification analysis of §8.1. Here we give the construction, which rests on two facts that coincide only in this setting:

1. **Stochastic retrieval is a randomized experiment.** Whether lesson v enters a run's context is the outcome of a stochastic softmax draw whose marginal inclusion propensity $\pi(v \mid c)$ is *recoverable from the logs* (we log the per-candidate scores and temperature and Monte-Carlo-estimate π; §5.2). Lesson inclusion is thus a stochastically assigned treatment with an estimable propensity—the textbook ingredient for causal identification, here approximate rather than closed-form.
2. **The reward is verifiable and re-checkable.** Re-running a solve and checking the flag costs compute but needs *no oracle*. So the counterfactual "reward without v" is directly measurable.

We therefore replace the confounded outcome signal with a *causal* credit signal—the lesson's marginal effect on reward—in three forms that trade cost against precision.

5.1 Active causal credit (leave-one-out re-rollout)

For each lesson v cited (executed) on a run, re-solve the same challenge from the same state with v *removed* from context, holding all other sampled lessons fixed, averaged over s independent re-rolls (the solver is stochastic; one

re-roll is a coin flip). Credit v with the marginal reward

$$\Delta R(v) \;=\; R_{\text{with}} \;-\; \tfrac{1}{s}\textstyle\sum_j R^{(j)}_{\text{without}}, \tag{4}$$

where $R = o\,(1{+}\lambda e)$ includes the step-count efficiency term, so a lesson the agent recovers from but is *slowed* by earns a negative marginal even when the outcome is unchanged (latent poison). This is precise but costs s extra solves per cited lesson; we gate it to cited lessons and report the re-roll budget.

5.2 Observational causal credit (inverse-propensity weighting)

The re-rolls are unnecessary when the data already contains the experiment. Over many runs, v is sometimes in context (treated) and sometimes not (control), with logged inclusion propensity $\pi_i(v)$ on run i. The marginal value of v is the *self-normalized inverse-propensity-weighted* (SNIPW) contrast

$$\widehat{\mathrm{ATE}}(v \mid c) = \frac{\sum_{i:\, v\in M_i} R_i/\pi_i(v)}{\sum_{i:\, v\in M_i} 1/\pi_i(v)} - \frac{\sum_{i:\, v\notin M_i} R_i/(1{-}\pi_i(v))}{\sum_{i:\, v\notin M_i} 1/(1{-}\pi_i(v))}, \tag{5}$$

estimated within each context c and shrunk toward the node's global estimate for sparse cells (partial pooling). $\pi_i(v)$—the marginal probability v is among the M sampled without replacement—is computed by Monte-Carlo from the logged per-candidate scores and temperature. This uses **no additional inference**: it reads the same logs the outcome bandit was fooled by. It recovers *large* marginal effects—the helpful lessons outcome credit conflates with poison—reliably; whether it recovers the *small* per-item harm of a benign-false lesson is exactly the identification question §8.1 answers in the negative.

5.3 The online observational bandit

Equation 5 yields a deployable system. The bandit samples lessons via the similarity-blended score as before, logging propensities; after each epoch it *recomputes* its value table $Q[c][v] \leftarrow \widehat{\mathrm{ATE}}(v \mid c)$ from its own accumulated logs and performs no per-run updates. The randomization that drives identification is the very exploration the bandit already does, so the credit signal is causal by construction at zero marginal cost. Active re-rolls (§5.1) then serve as an expensive but assumption-light validation of the cheap estimator: because they hold the other sampled lessons fixed, they are robust to the interference that the observational estimator must marginalize over, so agreement between them bounds the interference error.

Why this is the right tool for verifiable-reward memory. In an open-ended domain neither ingredient holds: lesson inclusion is rarely randomized with a known propensity, and the reward cannot be re-checked without an oracle. The verifiable-reward agentic-memory setting is special precisely because it supplies both—so the false-lesson problem that is intractable in general becomes, here, an estimation problem with a free experiment attached. "Estimable," though, is not "solved": §8.1 shows the per-item harm of a benign-false lesson is non-identifiable even with the experiment in hand, so the construction's value is the large helpful effects it does recover and the sharp boundary it lets us draw.

6 Experimental Setup

Why a controlled corpus. Live public CTF challenges confound two things we must separate: pretraining contamination (the model has seen the flag) and the absence of ground-truth causal labels (we cannot know *which* lesson caused a solve). We therefore build a corpus of small, automatically-verifiable tasks. Each task embeds a *freshly random* flag `picoCTF{···}`, so the model cannot have memorized it (zero contamination), and each family has a *known* correct method plus a planted *poison* lesson naming a plausible-but-wrong method. Descriptions are method-neutral, so a retrieved lesson—not the prompt—carries the method.

Calibrating the corpus to the solver (the key design step). A controlled corpus only measures false-lesson dynamics if it sits in the band where a lesson *matters*: too easy and the model solves regardless (a poison cannot hurt, a true lesson cannot help); too hard and it fails even with the true lesson. A first-pass corpus (§7) was bimodal for our 4B solver—trivial single decodes at ceiling, exact-crypto families at floor—so we recalibrate. Each family is now a two-layer decode in which an *outer* transport encoding (Base64 or hex, so `file` on the artifact reveals only "ASCII text") wraps an *inner* transform (gzip, raw zlib, single-byte XOR, or reverse) that is invisible in the raw bytes. The model can execute each step but cannot guess the inner transform from the artifact, so the retrieved *recipe* carries the method: the true lesson names the right inner transform; the poison names a plausible wrong one. The round budget is kept tight (four) so the model cannot simply explore every decoding. This yields the regime the study needs—the true lesson lifts solve sharply over no memory, and a poison co-retrieved with it is the clean instance of recovery masking. We use the families the 4B executes reliably once instructed (`b64_gzip`, `b64_rev`) plus harder zlib/XOR variants where it succeeds only sometimes, spanning an easy–hard difficulty range.

Model substitution (disclosed). The intended lab solver was Qwen3.5-27B-FP8 on dual A5000s (ultimately not run; the lab-scale ladder below uses Qwen2.5). The pilot runs on a single 8 GB laptop GPU, on which that model is infeasible; we substitute **Qwen3-4B-Instruct-2507** in 4-bit—a smaller model of the *same family* as the intended Qwen3.5 solver, one release apart—served

behind the same OpenAI-compatible API the solver already uses, with a small bi-encoder (all-MiniLM) as the retrieval embedder. The rerank stage in every reported run is a deterministic lexical-overlap-plus-cosine reranker, not a trained cross-encoder (which the code supports but we did not need): in the controlled corpus each family's true and poison lessons are on-target and dominate query similarity by construction, so which lessons reach the sampled M-set—and thus every logged propensity the causal estimators use—does not depend on the reranker's sophistication. The same-family choice is deliberate but limited: the spurious-reward effect is documented for the Qwen family specifically [35], and self-preference bias affects LLM-as-judge setups generally [24]; a Qwen-3-generation pilot shares the solver family but *not* the capability of the 27B lab model, and capability-dependent false-lesson dynamics may not transfer (§9). This is a capability substitution, not a method change; we report mechanism and lift. A calibration run placed the solver's no-memory solve rate near the middle of the range (54%), giving headroom for lessons to help *and* for poison to harm—the regime the study needs.

The lab-scale ladder (where the headline lives). The 4B pilot *illustrates* the boundary and the credit-signal failure mode; the load-bearing non-identifiability evidence—the active leave-one-out gold standard (§8), the positive control, the capability ladder, and the base-rate floor of §8.1—comes from a separate set of *lab-scale* campaigns that run the same instrument and pipeline at M=3. The ladder is the **Qwen2.5-Instruct** family—3B, 7B, 14B-AWQ, 32B-AWQ—plus a non-Qwen rung (**Llama-3.1-8B-Instruct**); the rungs log 2.2k–15.6k runs over 11–43 challenge families each (∼15.6k/15.5k at the powered 14B/32B rungs, over 41/43 families), and a malignant positive-control campaign (14B, 4264 runs, later expanded by a further 5096) adds the actively-destructive lesson types. We are explicit that this ladder is a *different model generation* from the Qwen3-4B pilot (and from the intended-but-unrun Qwen3.5-27B lab solver): the cross-rung consistency we report is therefore *within* the Qwen2.5 family plus one Llama rung, not across the pilot's generation, and "across architecture" means the single Llama rung (other non-Qwen probes were harness-floored at $n \leq 3$; §9).

Baselines and ablations. **B0**: no graph (the lift denominator). **B2**: static similarity retrieval with reranking and diversity, no bandit. **B4**: the full method (bandit, cited-only credit, step reward). We ablate the two oracle-free levers: *all-M credit* (credit every in-context lesson, not just cited ones) isolates recovery masking, and *outcome-only reward* (λ=0) isolates latent poison. **B5/B6**: gated vs. ungated write-back. Each learning system streams the corpus for two epochs; B4 is then frozen and evaluated paired against B0/B2 on common random seeds.

Metrics and statistics. Solve rate and paired lift over B0 with bootstrap confidence intervals and exact McNemar tests [7, 21]; mean learned Q by ground-truth lesson label; poison citation rate and recovery-masking rate; and the leave-one-out marginal value (false-lesson exposure). Difficulty is ground-truth (family point value); the bandit budget calculation is reported.

7 Results: The Boundary

We first report a pilot on an initial micro-CTF corpus. It establishes the boundary the rest of the paper addresses and, in doing so, exposes the benchmark-calibration lesson that motivates the recalibrated corpus (§6) used for the causal-credit study in §8. We report findings in four parts. All paired comparisons use common random seeds and bootstrap confidence intervals; we report the headline comparison at the *challenge* level (independent units) because our 80 rows are 40 challenges × two seeds, and the seeds are highly correlated (identical B0 outcome on 37/40 challenges).

(1) Retrieval helps, but the lift is concentrated. Static retrieval (B2) reaches 66% vs. 54% with no memory—a paired lift of +0.12 that remains significant at the challenge level ($p = 0.031$, 40 challenges, a challenge counted solved if solved under either seed: exact McNemar over the six discordant challenges, all favoring B2; Table 1, Fig. 2). The lift is *not* uniform: 5 of ten families are already solved with or without memory and 2 are unsolved in both conditions, so the entire effect comes from 3 families (ROT13, reverse, and the two-layer Base64∘ROT13). The lessons carry value precisely where the model is neither saturated nor blocked. This concentration is itself the reason false-lesson propagation is worth studying here: a memory that helps on a few families can still accrue incorrect lessons that the aggregate solve rate hides. (The methodological note stands regardless: the prior between-subjects evaluation reported a comparable effect that was not significant; the paired design resolves it.)

(2) Action-grounded credit zeroes *unused* false lessons. Cited-only credit assigns no update to a lesson that emitted no command, so unused (distractor) lessons stay at the prior, $Q = +0.00$, whereas all-M credit rewards them on every success, $Q = +0.36$ (Fig. 3). This zeroing follows directly from the update rule; the *empirical* content is the magnitude of spurious credit it withholds—a 69% recovery-masking rate, i.e. that fraction of in-context lessons present during a success were never executed and would have been credited by all-M. So the lever genuinely matters for the *passive* mode, but its effect is mechanical, not learned.

(3) Executed poison evades the verifiable signal. When the false lesson is itself executed, cited-only credit

System	Solve rate	Lift vs B0	95% CI
B0 (no graph)	0.54	—	—
B2 (static retrieval)	0.66	+0.12	[+0.05,+0.21]
B4 (bandit, full)	0.61	+0.07	[+0.00,+0.15]

Config	Q(true)	Q(poison)	Q(distractor)
cited-only + step (full)	+0.34	+0.40	+0.00
all-M credit	+0.29	+0.29	+0.36
outcome-only reward	+0.37	+0.40	+0.00

Table 1: Top: solve rate and challenge-level paired lift over no memory (B4 is the frozen bandit and does *not* beat B2). Bottom: final learned value Q of true, poison, and unused (distractor) lessons under each configuration. The bandit values executed poison indistinguishably from true lessons at this n.

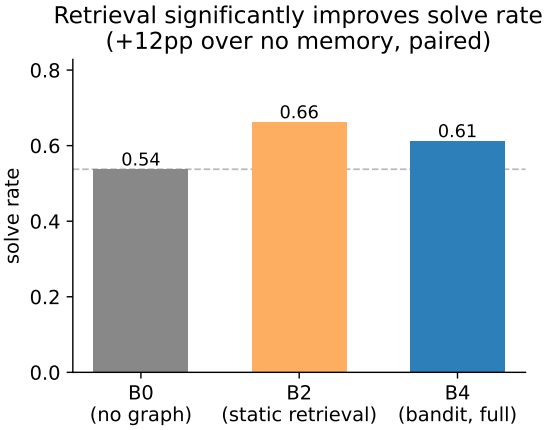


Figure 2: No memory (B0) vs. static retrieval (B2) vs. the frozen bandit (B4). Retrieval lifts solve rate; the bandit (B4) does *not* improve on static retrieval at this scale—its interest is in *what it learns about lessons*.

cannot help. Poison lessons name a wrong tool the agent runs (cited on 44% of the runs they are sampled in); on a *recovered* success they receive the same positive credit as a useful lesson. The bandit therefore values poison indistinguishably from true lessons ($Q_{\text{poison}} = +0.40$, in this pilot corpus numerically above $Q_{\text{true}} = +0.34$; Fig. 4), and the frozen bandit does not beat static retrieval (61%). The step-count lever does not change this ($Q_{\text{poison}} = +0.40$ under outcome-only reward as well)—but for a reason that is itself a limitation: this corpus's poison misleads via the *wrong tool*, not a longer route, so the latent-poison mode the step-count reward targets is never instantiated here, and that lever is *specified but not empirically tested* (§9). The clean statement is: a verifiable success credits whichever tool was run, true or false; the verifiable signal alone, even with action-grounded credit, does not attribute cause.

(4) An oracle-free causal probe measures marginal necessity. The leave-one-out re-rollout probe re-solves a challenge with and without a single lesson, from a shared

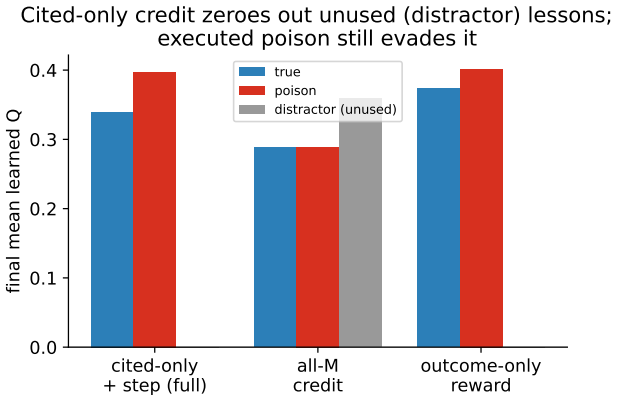


Figure 3: Final learned value by ground-truth lesson type. Cited-only credit holds *unused* (distractor) lessons at zero; all-M credit rewards them. But *executed* poison keeps a value indistinguishable from true lessons under every configuration (n=4/label)—outcome credit cannot tell them apart.

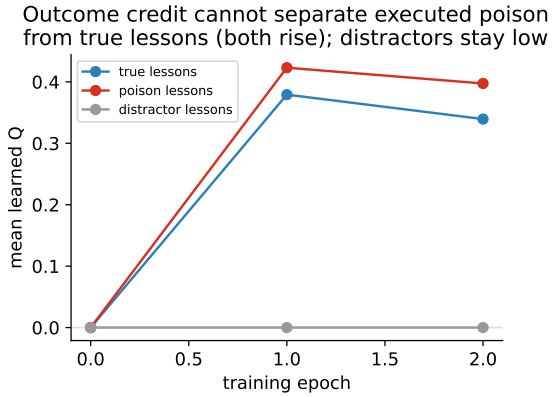


Figure 4: Learned value over training (full method). Executed poison tracks true lessons—outcome credit cannot separate them—while unused distractors stay at zero.

state, using only the flag. It measures single-lesson *necessity*; whether it can *rank* the small per-item harm of a benign-false lesson is a separate question that the lab-scale battery answers in the negative (§8). On the ten probed challenges—one true and one poison lesson each, twenty challenge–lesson pairs (drawn from *five* of the ten families: Base64, hex, ROT13, reverse, gzip—the python-scripted families XOR and Caesar, and the largest lift driver Base64∘ROT13, are *not* probed)—poison lessons have marginal contribution $\Delta\text{solve} = +0.00$, while true lessons have $= +0.40$. That mean is bimodal: true lessons are causally *necessary* ($\Delta\text{solve} = +1.0$) only on the families the model cannot solve unaided (ROT13, reverse) and 0 where it already succeeds—so the probe measures genuine necessity, not correlation with success. Two caveats are essential and we state them plainly. First, the probe and the bandit Q are different estimands (single-lesson necessity vs. average reward over an M=8 context), so part of the apparent "probe succeeds where Q fails" gap

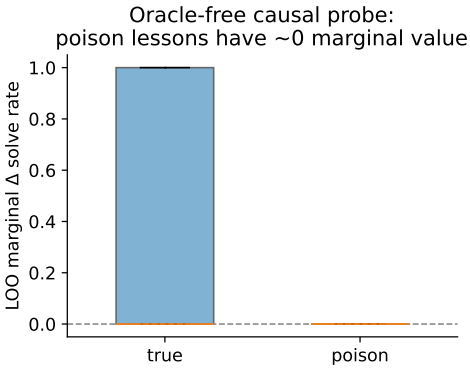


Figure 5: Leave-one-out marginal contribution, by ground-truth lesson type, on the five LOO-probed families. Poison shows $\Delta \approx 0$ here—partly true by construction (a non-reaching wrong tool), and its small *per-item* effect is not rankable at scale (§8)—where the bandit's outcome-credit had valued it like a true lesson; true lessons are causal only where the model cannot already solve unaided.

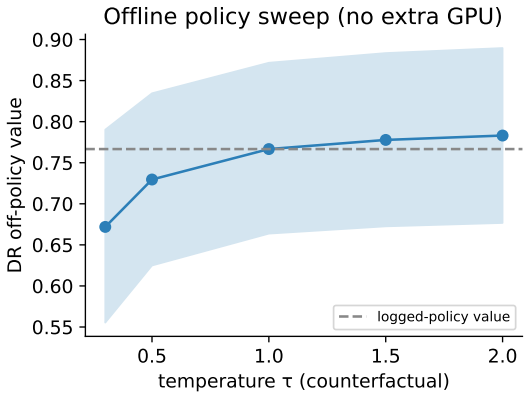


Figure 6: Off-policy value of counterfactual sampling temperatures from logged propensities. The doubly-robust estimate recovers the logging policy's measured value (dashed); the top candidate is within its CI, so the sweep validates the estimator without identifying a significantly better policy at this n.

is the change of estimand; second, because our poison is a non-reaching wrong tool, its non-causality is *partly true by construction*—the stronger test has two halves: poison that reaches the flag by a longer route, which we construct in Appendix C and probe against the *step* channel only (the LOO probe was not run against it), and LOO at matched M=8 conditioning, which remains future work. The two remaining levers behave *as specified*: gated write-back leaves 72% of new lessons (13/18) provisional and trusts only 5/18, against 13/19 (68%) trusted under ungating—though we report the provisional/trusted split, not whether the trusted nodes were causally correct. Finally, off-policy evaluation recovers the logging policy's measured value (doubly-robust 0.77 vs. measured 0.77); this sanity check is exact because the importance weights are unity when $\pi' = \pi_{\log}$, so it *validates the estimator* rather than identifying a better policy—the top counterfactual temperature ($\tau = 2.0$) lies within the logging policy's CI (Fig. 6). Validating under genuine distribution shift requires a second on-GPU batch and is future work.

Summary. The verifiable flag, with action-grounded credit, cheaply removes value from *unused* false lessons and (with gating) withholds trust from *premature* ones, but *cannot* down-weight executed false lessons the agent recovers from—the engineered bandit therefore does not improve on static retrieval here. A causal re-rollout measures single-lesson necessity on the subset we probe—and §8 pushes this to its limit, showing that even this gold standard cannot rank the per-item harm of a benign-false lesson. This maps which false-lesson modes the verifiable signal alone can contain, and where causal credit reaches and where it does not.

8 Results: What Verifiable Credit Can and Cannot Identify

We instantiate the credit signals on the recalibrated corpus (§6): **B0** (no memory), **B2** (static similarity retrieval), **outcome** (the bandit with outcome credit), and **observational** (the online observational causal bandit, §5.3); an active leave-one-out re-rollout (§5.1) serves as the assumption-light *gold standard* against which we judge the cheap signals. Each learning condition streams the 24 training challenges for 5 epochs, is then frozen, and is evaluated paired against the others on the 8 held-out instances over 3 seeds. The held-out set and seeds are identical across conditions, so every contrast below is within-item paired (bootstrap CI + exact McNemar). Rows are instance×seed and seeds are positively correlated, so these campaign p-values are anti-conservative; no campaign-level contrast is claimed significant. The gold-standard re-rollout, the positive control, and the capability ladder are established at *lab scale* on the same instrument (the high-n campaigns, e.g. `campaign_14b_M3_powered`, $n \approx 15.6$k); we say so at each use. One provenance caveat: the 4B campaign itself ran on an earlier benchmark revision (poison-contract / oracle-labeling / grid-duplication bugs fixed later in commit `63db02d`) and was *not* re-run, so the 4B magnitudes below are illustrative of the failure mode—the load-bearing non-identifiability evidence is the lab-scale ladder, not the 4B numbers.

The boundary, reproduced on real logs. Outcome credit cannot tell an executed-but-recovered-from poison lesson from a useful one. Over the full training stream it learns a mean Q of +0.283 for true lessons and +0.183 for poison: it ranks true lessons only slightly higher (gap +0.100, n=4/label—too few for a formal test) and assigns poison a clearly *positive* value, because a verifiable success credits whichever lesson was in context and the poison's

claimed action (`base64`) is part of the correct recipe it cannot be denied credit for. An outcome-credit memory therefore holds no internal signal that executed poison is anything but a useful lesson—the failure mode this paper is about.

The free experiment identifies the *helpful* lessons. The same logs contain a free randomized experiment: each lesson's inclusion is a stochastic draw with a logged propensity (§5.2). Re-scoring *those same* outcome-condition runs with the observational SNIPW estimator (§5.2)—no extra inference—recovers the value of the helpful lessons outcome credit conflates with poison: the mean IPW ATE on reward is +0.158 for true lessons, a large positive effect, with distractors at -0.053. This large helpful effect is where the agent's own exploration *does* suffice to credit a lesson causally for free, and it is corroborated by the gold standard: at lab scale the active re-rollout ranks the helpful lessons in strong agreement with the cheap estimate (within-helpful rank correlation +0.64, CI $[+0.34, +0.85]$; the gold standard orders 30/30 helpful nodes)[2] and the helpful on-target solve ATE (the *on-target* contrast restricts the run set to challenges of the lesson's own family—family is pre-treatment—as opposed to the *marginal-inclusion* effect of merely placing it in the retrieved M-set) is large at every rung and markedly larger at the high-capability Qwen rungs (+0.247/+0.195/+0.526/+0.570/+0.227 at 3B/7B/14B/32B/Llama-8B, all $p < .001$; §8.1). This is the *positive* half of the boundary.

. . . but not the per-item harm of a poison lesson. The negative half is the paper's central result. The same observational estimator reads poison only *mildly* negative—-0.104 on this 4B campaign, and at higher capability with far more data a small −0.057 ($p<.001$, CI $[-0.087, -0.022]$) at 14B and −0.064 ($p=.005$, CI $[-0.109, -0.018]$) at 32B—the pre-registered per-rung primary contrast at each rung; the 14B/32B cells survive Benjamini–Hochberg at $q=.05$ across the five rungs, the other rungs are null (all other grid cells are exploratory). These are the *on-target* contrasts; the marginal-inclusion ATE is null at every M (§9).[3] That sign is *not* the per-item causal harm (Fig. 7). Against the assumption-light active leave-one-out gold standard—which re-solves each challenge with and without a single lesson, holding the others fixed—the cheap estimate cannot *rank* the per-item poison effect: within poison, the rank correlation between the cheap estimate and the gold standard is ≈ 0 (+0.072 tie-corrected Spearman, +0.028 rank-Pearson; wide bootstrap CI $[-0.35, +0.49]$, emitter-backed, seed 0), and among the lessons the gold standard can order at all, the two agree on the *sign* of only 8 of 18 ($p=0.76$, one-sided exact binomial against chance 0.5—at chance, not anti-correlated).[4] The reason is not merely that the cheap estimator is noisy: the feasible-n gold standard *itself* cannot resolve the small poison effects. Its mean poison marginal is +0.045 ($n=28$ lessons), 95% CI $[-0.006, +0.098]$—a fail-to-reject, *ill-posed*, never a measured "zero"—and its per-node standard error (a worst-case $p=0.5$ binomial bound at the deployed re-roll budget, ≈ 0.20) exceeds the between-node spread of the effects it must rank (0.14). **Neither** signal identifies the per-item harm—and if anything this understates the difficulty: the active-LOO contexts themselves rendered the poison lesson *first*, biasing its per-item effect toward looking *more* harmful, so the gold standard's inability to rank it is conservative. The mildly-negative observational sign is a real *label-level* regularity but a confounded one—§8.1 derives its source under the measured base-rate floor—and it is neither the per-item causal estimand nor, as we show below, a signal a write-back gate can be built on.

A positive control: what verifiable credit *can* catch. The non-identifiability is specific to a *benign*-false lesson the agent recovers from, not a blanket failure of the estimator. We additionally plant two *actively destructive* lesson types that push the outcome *below* the base-rate floor—one that destroys working state (`destroy`) and one that burns the round budget (`exhaust`)—a `wrongflag` lesson that confidently submits a wrong answer, and a deterministic `verifier` that re-runs the cited recipe in isolation. The SNIPW estimator *detects* the floor-violating harms (`destroy` ATE −0.016, two-sided $p=0.010$, CI $[-0.028, -0.004]$, 74 families; `exhaust` −0.017, $p=0.017$, CI $[-0.030, -0.003]$, 71 families; both CIs excluding 0, though the binary `destroy` clears $p < .05$ only once campaigns are pooled—see note) but is *untested* on `wrongflag`. The agent executed that recipe on only 11 of 453 sampled runs (follow rate 0.024, against 0.978 and 0.805 for `destroy` and `exhaust`), so its on-target SNIPW (+0.004, $p=.69$, 64 families—the same pooled estimator and file as `destroy`/`exhaust` above, `malignant_pooled_combined.json`) carries almost no treated exposure: this is a null from non-exposure, not a demonstrated blind spot. Under *forced* exposure the constructed leave-one-out does detect the arm (−0.43, CI $[-0.54, -0.32]$, $n=41$ families; direction and sign only—see note).[5] The execution-verifier flags invalid recipes on 100% and valid ones on 0% of families *by construction* (a deployable contrast on *recipe validity*, not a measured generalization; §8.1, Prop. 4). The malignant campaign over-samples its planted lessons

[2] Lab-scale, `campaign_14b_M3_powered`; emitted by `experiments/within_poison_validation.py` into `active_loo_within_label.json`.

[3] Lab-scale on-target SNIPW, `campaign_14b_M3_powered` and `campaign_32b_M3_tp2` (§8.1).

[4] `experiments/within_poison_validation.py`; the often-quoted "8/28" counts 10 gold-standard ties-at-zero (a 1/12-quantized estimate) as disagreements.

[5] The `wrongflag` arm was already toothless at the pre-registered probe—cited on 0 of 42 sampled runs (`campaign_14b_malig_probe/merged/malignant_probe_check.json`)—which the pre-registration required be redesigned before scaling. It was scaled unmodified; we log this as a deviation and treat the arm as untested rather than as evidence of a blind spot.

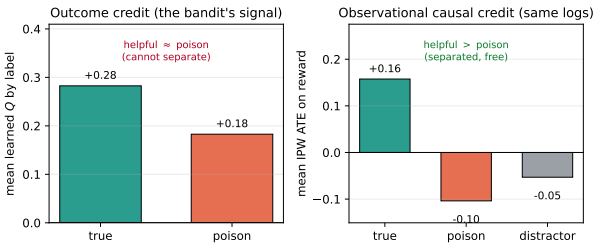


Figure 7: Same logs, two credit signals (4B recalibrated campaign, run on the earlier pre-fix benchmark revision; magnitudes illustrative). *Left:* outcome-bandit Q by ground-truth label—executed poison earns a positive value, numerically beside true lessons. *Right:* observational IPW ATE on the same runs—poison mildly negative, true strongly positive. The mild negative is a confounded label-level sign, *not* a per-item poison detector (§8.1).

condition	held-out solve
B0 (no memory)	29.2%
B2 (static)	45.8%
outcome credit	50.0%
observational causal	41.7%
paired contrast	**Δ (McNemar p)**
B2 − B0	+16.7 pp ($p = 0.289$)
observational − outcome	-8.3 pp ($p = 0.500$)
observational − B2	-4.2 pp ($p = 1.000$)

Table 2: Top: held-out solve rate by credit signal (4B campaign, run on the pre-fix benchmark revision and not re-run—illustrative; see the provenance caveat opening §8). Bottom: within-item paired contrasts (McNemar). Retrieval lifts solve over no memory, but the credit signals are not separated on solve rate at this n (an underpowered fail-to-reject, not a measured equality; both paired intervals are bounded above by 0.0) (and, as the text shows, in any trustworthy per-item poison value).

∼2.5×, which depresses the base rate and makes the *same-campaign* benign poison read spuriously positive (run-level +0.033, one-sided p=.0475, two-sided .095)—so the benign-poison null is the *main*-campaign result, not this one—but destroy/exhaust are floor-*violating* and detected regardless.[6] This is the boundary in one experiment: outcome-based credit identifies large helpful effects and actively-destructive harm, but *not* the per-item harm of a benign-false lesson, and it is *untested* against a confidently-wrong answer.

Held-out solve rate (parity, by design). Table 2 reports solve rates; the headline is *not* here, and the next paragraph explains why it should not be. Retrieval is directionally ahead of no memory (B2 45.8% vs. B0 29.2%, paired +16.7 pp, $p = 0.289$—underpowered here; the pilot's paired lift, §7, is the significant version of this contrast), consistent with the recalibrated corpus putting the solver in a band where lessons matter. But the credit signals are not separated at this n: the online observational causal bandit reaches 41.7% against the outcome bandit's 50.0% (paired -8.3 pp, McNemar $p = 0.500$) and static retrieval's 45.8% (paired -4.2 pp). Whatever the credit signal, the bandit retrieves the true lesson and the solver follows it, so solve rate does not separate the signals—and, as we have shown, neither does any of them recover the per-item poison effect.

Where the harm does and does not surface. The credit signals are at parity on *immediate* held-out solve rate, expected on the *marginal* retrieval distribution: a capable solver follows the true lesson whenever it is co-retrieved and ignores a co-retrieved false one, so a false lesson does not lower immediate solve in single-step retrieval. Two qualifications keep this honest. First, under *forced* co-presence in a tight round budget, adding a poison lesson alongside the true one *does* lower solve, by ≈0.31—but this effect is *not* a per-item constant (poison alone is mildly recoverable-*positive*, +0.05, CI excluding 0, floor-bounded at base ≈0.04; difference-in-differences −0.37) and it is resource/co-presence-mediated: against a pure budget-burner (exhaust) control, poison hurts *no more*—poison−exhaust is +0.042 (p=.78) on the solvable subset (a cannot-reject-equality, underpowered at n=15—*not* a formal equivalence (TOST) test) and +0.204 ([+0.040, +0.357], p=.0164) over all 41 families (poison hurts *less* than the maximal burner). Disclosed: those constructed rollouts rendered the poison lesson *first* (a primacy confound, since fixed), so we do not headline the magnitude.[7] The reading consistent with the gold standard is that there is no clean, order-robust *per-item* poison value—further support for the non-identifiability result, not a separate harm. Second, the small negative is an *on-target* (family-restricted) contrast, *not* the marginal-inclusion ATE: conditioned on a poison lesson actually being executed, the solve rate dips a measurable −0.0745 ($p < .001$) at full exposure (M=1) within crypto, while the causal marginal-inclusion ATE stays null at every breadth (M=1 −0.021, p=.25). We treat this conditional sub-floor harm as the strongest rival reading, which we name in §9.

Does the confounded sign make a usable gate? No. It is tempting to read the observational poison-negative as a write-back filter. Applied as a promote-if-positive gate to

[6] Lab-scale malignant campaign, family-clustered SNIPW *pooled* over `campaign_14b_malig` + `campaign_14b_malig_expand` (74/71 family clusters in the destroy/exhaust contrasts); the single-campaign binary destroy was borderline (p=.057) and the campaign was expanded to power it, so the robust single-campaign evidence is the descriptive within-family destroy ATE −0.045, CI [−0.055, −0.032] (excludes 0 without pooling). The constructed direct-harm magnitudes are forced-exposure and order-confounded, so they are direction/sign evidence, not clean effect sizes.

[7] `experiments/crypto_interference_analysis.py` over `constructed_loo_crypto.json`; the order-neutral renderer (`solver/context_assembly.py`, content-hash order) is the fix.

each signal's learned values on the *same* outcome-condition logs, it would admit 100% of the planted poison under outcome credit and 0% under the observational value. We report this only to *disavow* it as a deployable result: the gate "works" here solely because the observational sign happens to be negative on these four lessons, and that sign is exactly the confounded, non-identifiable quantity the gold standard cannot reproduce (within-poison rank correlation ≈ 0); the 0% figure is itself fragile—the same observational estimator admits 25% of the poison when read off the *observational*-condition logs rather than these outcome-condition ones—and the lab-scale causal gate is only *partially* label-selective, at a real cost: at 14B it promotes 29/40 of the helpful nodes but only 13/39 of the poison (Fisher p=6.8×10^{-4}), and 29/42 against 15/42 at 32B (p=.004), while at 7B it does not separate the labels at all (18/35 vs. 16/33, p=1.0; likewise null at 3B, 6/12 vs. 4/12, and Llama-8B, 5/11 vs. 3/11). For comparison the outcome gate promotes essentially everything (40/40 helpful and 37/40 poison at 14B). That separation is the same confounded *label-level* regularity, not the per-item ranking a gate needs—and buying it costs between a quarter and a half of the helpful nodes at the two selective rungs, and about half at the three where it does not separate. With four lessons and a sign we have shown is *not* the causal estimand, this is an illustration of the failure mode, not a working suppressor. A trustworthy gate would need a signal that ranks the per-item poison effect, which—in this regime—neither the cheap estimator nor the gold standard provides. Whether *any* oracle-free signal can gate benign-false lessons, and the downstream cross-challenge solve-rate cost of propagated poison at scale, is the open problem we name in §9.

8.1 When can outcomes isolate a harmful lesson? An identifiability result

The causal-credit estimators of §5.2–§5.3 measure each lesson's marginal effect on reward. A natural hope is that the *same* signal that surfaces a genuinely useful lesson will, with the sign flipped, surface a *benign-harmful* one—a lesson that is causally wrong but not actively destructive (our "poison": it names a wrong tool the agent runs and then recovers from). §8 shows the outcome signal does not: helpful lessons are detected at every capability rung, while on the binary solve the poison effect is a clean null at 3B/7B/Llama-8B and only a small negative at 14B/32B (mildly *below* the floor at 32B) that the assumption-light leave-one-out gold standard cannot rank. This subsection explains *why*, under a model whose premises we measure directly and report honestly where the measurement only *approximately* holds. The content is *not* the trivial algebra that a floor-equalized contrast is zero—that is one line—but its consequence for an entire *class* of estimators: under the measured floor every consistent outcome-based inverse-propensity/AIPW estimator converges to zero against on-target poison—any test built on one has trivial asymptotic power—so the binary outcome cannot identify it. (A naive "exact" conditional-randomization test does not rescue this: as implemented it is *anti-conservative* for top-m retrieval and *blind* on our positive control, so it is non-diagnostic here; we demote it to a one-line disclosure below.) We then examine the reward channel and a direct execution observable that restores a consistent estimator for a related, recoverable target. This is an *identifiability/detectability-power* statement under a stated, empirically checked (at 32B only approximately satisfied) model—a domain-specific identification boundary, *not* a universal impossibility, and *not* a "harm vanishes as the base rate vanishes" law (which our data do not support at the powered rungs; see "Reconcile with data", where we also report the one rung that runs the other way).

Setup and symbols. Fix a challenge *family* f (a transport-encoded challenge class; §4) and a retrieval context c (the bandit's coarsened key, a category/cluster). Let the binary outcome $o \in \{0,1\}$ be a verifiable solve and $R = o\,(1{+}\lambda e)$ the step-efficiency reward of §5.2, with e the standardized step-count term. Define the family *base solve rate* $b_f = \Pr(o = 1 \mid f,\ \text{no on-target recipe cited})$; empirically we estimate b_f per family as the solve rate of runs in which *no* on-target recipe is cited (the arm-0 rate of the A1 contrast); the off-target-*cited* solve rate is close but not identical (gaps up to ≈0.03–0.06 depending on rung and weighting), and we use b_f throughout. For an *on-target* helpful lesson v (recipe matches family f) write $h_f = \Pr(o = 1 \mid f, v \text{ helpful, cited})$. As in §5.2, $\pi_i(v)$ is the logged marginal inclusion propensity of v on run i (Monte-Carlo from the softmax scores), M_i the sampled set ($|M|$=3 in the lab campaigns), and $\widehat{\mathrm{ATE}}(v \mid c)$ the SNIPW/IPW contrast of Eq. (5) with $\mathrm{ATE}(v \mid f)$ its population estimand: the per-lesson average treatment effect of *including* v in the retrieved set, on a target (o or R), within family f. Two variants appear below and *neither* conditions on citation: the *on-target* contrast restricts the run set to challenges of v's own family (family is pre-treatment, so this is a conditional ATE, not a selection effect), while the *marginal-inclusion* contrast pools over all families—over the off-target retrievals a deployed memory also makes. The shift between them is by family-stratification of the same logs. The one citation-conditioned statistic in this paper is A1's own contrast (poison-cited vs. no-recipe), a different estimator. "Cited" means the lesson's tool/technique signature appears in the executed trajectory (action-grounded; §4).

The model: two premises, measured (A1 exact at three rungs, above-floor at 3B, approximate at 32B).

Assumption 1 (Outcome-level base-rate floor for benign-wrong lessons)**.** *When an on-target* poison *lesson is cited, it contributes no usable progress* to the outcome*:*

the run's solve-probability collapses to the no-recipe one,

$$\Pr(o = 1 \mid f,\ v \text{ poison, cited}) \ = \ b_f.$$

Executing a benign-wrong recipe is, for the binary outcome*, equivalent to executing no recipe—the agent falls back on its own priors and recovers (or not) at the base rate. This constrains o only, not the step count (see Hypothesis 1).* Empirical check (family-stratified contrast, all five rungs). *The declared estimator is the within-family contrast poison-cited$-b_f$. It includes* 0 *at three rungs—7B* −0.001*, CI* [−0.022, +0.024]*; 14B* −0.012*, CI* [−0.029, +0.007]*; Llama-8B* +0.014*, CI* [−0.008, +0.040]*—so A1 is not distinguishable from exact there (point estimates 7B* −0.001*, 14B* −0.012*, Llama-8B* +0.014*). At 3B it does* not*: the contrast is* +0.036*, CI* [+0.002, +0.077]*, a small deviation* above *the floor. We state this plainly because it cuts* for *the result rather than against it: an above-floor deviation is not hidden harm—it leaves $s_f = 0$ in Prop. 3, and a consistent outcome estimator would converge at 3B to a small* positive*, ranking cited poison mildly* helpful*, which sharpens rather than weakens the non-identification. (The unstratified marginals agree the rates are close—3B* 0.097 *vs.* 0.072*; 7B* 0.091 *vs.* 0.099*; 14B* 0.194 *vs.* 0.195*; Llama-8B* 0.077 *vs.* 0.064*—but the stratified contrast is the statistic we report.) At 32B A1 is* mildly violated in the harmful direction*: the contrast is* −0.040*, CI* [−0.062, −0.016] *(excludes* 0*; n*=38 *families)—a small* sub-floor *harm of* $\leq \sim$4*pp, the same* on-target *(family-restricted) phenomenon as the full-exposure (M*=1*) on-target sub-floor of §8 (*−0.0745*, $p < .001$; the marginal-inclusion ATE stays null at every M). We therefore restate A1 as* exact at deployed M for 7B/14B/Llama-8B, above-floor at 3B, and approximate (|sub-floor| ≤ 4pp) at 32B*, and read the result's exact-zero statement as the small-sub-floor limit, not a claim of "no sub-floor harm anywhere."*[8]

Assumption 2 (Near single-item retrieval: little helpful mass to displace)**.** *In the analyzed regime, on-target citation is effectively single-item per family: when poison is cited it rarely crowds a* helpful *on-target lesson out of the executed context within the same run. Formally the helpful-displacement probability δ_f =* Pr(*a helpful on-target lesson is displaced | v poison cited, f*) *is small. We treat $\delta_f \to 0$ as the* limit *in which the result's exact-zero statement holds; the finite-δ_f residue is governed by Prop. 3, not by Prop. 2.* Empirical check (and caveat): *the campaigns sample |M|*=3*, so single-item-ness is not automatic; its justification is empirical: although each family plants both a helpful and a poison on-target lesson, the two are rarely cited in the same run (measured co-citation* 0.04–0.13 *across rungs), so on-target helpful and on-target poison rarely co-occur for the* same *f. The directly-measured co-citation displacement is small at every rung ($\widehat{\delta}_f$ -0.020/+0.042/+0.062/+0.050/+0.060 at 3B/7B/14B/32B/Llama-8B; negative at 3B, i.e. displacement in the other direction), consistent with near single-item retrieval; the decomposition below uses the 14B and 32B values; we use this* measured $\widehat{\delta}_f$ *(not a back-fit) in the decomposition below. The high follow/cite rates (*0.76–1.0 *across rungs and labels, mostly* ≥0.90*) are necessary but not by themselves sufficient evidence for small δ_f; the co-citation rate is the quantity that directly bounds it.*

A1 is the load-bearing, separately measured premise; A2 delimits scope. Under genuinely multi-item retrieval poison *can* harm, but only by displacing helpful mass, and that harm is bounded (Prop. 3); A2 is the small-δ_f limit of that channel. We do *not* assume the reward is floor-equalized: the step channel is handled explicitly in Hypothesis 1 and Prop. 3.

Identification of helpful lessons.

Proposition 1 (Helpful lessons are identified by outcomes)**.** *For an on-target helpful lesson,* $\mathrm{ATE}(v \mid f) = h_f - b_f > 0$ *whenever $h_f > b_f$. Under propensities bounded into $[\epsilon, 1-\epsilon]$ in the analyzed cells (overlap, enforced by the exploration floor and the ESS/clip guards of §5.2),* $\widehat{\mathrm{ATE}}(v \mid c)$ *is consistent for this contrast, which is large and significant in the data (§8:* +0.195 *to* +0.570 *across rungs, all $p < .001$).*

Proof sketch. Citation is a stochastically assigned treatment with known propensity $\pi_i(v)$, so the SNIPW contrast of Eq. (5) is a Hájek estimator of the treated/control mean difference, consistent under overlap and bounded R; under A2 its estimand on o is $h_f - b_f$. When $h_f > b_f$ the treated reward law is stochastically distinct from control, so the SNIPW contrast separates from 0 at a rate growing with the run count. □

Non-identification of benign-harmful lessons from the outcome. We first isolate the trivial algebra, then state the substantive (estimator-class) content.

Lemma 1 (Floor-equalized outcome contrast)**.** *Under A1–A2, the population on-target poison effect* on the binary outcome *is exactly zero:* $\mathrm{ATE}_o(v \mid f) = b_f - b_f = 0$.

Proposition 2 (Outcome powerlessness against benign-wrong lessons, under A1 (exact at 7B/14B/Llama-8B, above-floor at 3B, approximate at 32B)–A2)**.** *Under A1–A2, the treated and control* outcome *laws coincide, $o \perp\!\!\!\perp \mathbf{1}\{v \in M\} \mid f$. Consequently* every *consistent outcome-based estimator—SNIPW or AIPW/DR keyed on o—converges to* 0 *(Lemma 1): no outcome-based procedure can distinguish a benign-wrong lesson from an inert one. Equivalently, the harm of a benign-wrong on-target lesson is* not identifiable from the binary verifiable outcome of

[8] Cite rates and the *A*1 contrast trace to the high-*n* powered campaigns (`campaign_14b_M3_powered` and `campaign_32b_M3_tp2`, `merged/a1_contrast.json`, $n \approx$15.6k at 14B); the ladder ATEs below are from the same campaigns.

single trajectories. *The substantive content is this powerlessness of the whole estimator class, not the one-line zeroing of the contrast. (An idealized exact randomization test keyed on o would likewise have uniform p under this law-equality; we do* not *lean on it, because the test* as implemented *is anti-conservative for top-m sampling-without-replacement—see the CRT disclosure under "Reconcile with data.")*

Proof sketch. By A1 the cited-poison outcome distribution equals the no-recipe one (A1 governs *cited* poison; a sampled-but-uncited poison executes no recipe and so also follows the no-recipe law, and on-target cite rates are 0.76–0.99, so the cited/included gap carries little outcome mass); by A2 the control side carries no displaced helpful mass, so the treated and control outcome laws are identical, $o \perp\!\!\!\perp \mathbf{1}\{v \in M\} \mid f$. Any consistent estimator of a functional of two coinciding laws converges to that functional at equality, namely 0 (Lemma 1); the SNIPW/AIPW contrasts on o are such functionals. The only realizable outcome-level poison effect is the displacement term of Prop. 3, which A2 sends to 0, plus the small measured sub-floor residue at 32B (A1). *This result governs o only*; the reward channel is Prop. 3/Hypothesis 1. □

The reward channel is not a clean null: a step-detour penalty. A1 equalizes the outcome but says nothing about *steps*. Because $R = o(1+\lambda e)$ also reads the standardized step count e, the reward can carry signal the outcome cannot.

Hypothesis 1 (Conditional step law—not a measured premise)**.** *One might hope the cited-poison step-count law differs from the no-recipe law (a detour through the wrong tool costing steps), which would let the* reward $R = o(1+\lambda e)$ *carry signal the binary outcome cannot. We flag this only as a* hypothesis*; the directly measured contrast does* not *support it.* Empirical check (against the hypothesis): *the on-target poison* reward *ATE is* 1.7–2.1× *the outcome ATE (14B* −0.119*; 32B* −0.106*, vs. outcome* −0.057/−0.064*), but this ratio is an* R-scale identity*, not a treated-side detour—the same ratio holds for* helpful *lessons (14B* 2.12 *vs poison* 2.09*) and reflects R's amplification of the displacement of* fast *helpful solves. The direct conditional step contrast (solved runs, poison-cited vs no-recipe, family-weighted) does not support it either, but we report all five rungs because they do not agree:* +0.064 [−0.350, +0.517] *(3B),* +0.182 [−0.213, +0.601] *(7B),* −0.227 [−0.407, −0.056] *(14B,* reversed*—poison-cited solves are faster, the one rung whose CI excludes* 0*),* +0.010 [−0.122, +0.154] *(32B),* +0.340 [−0.119, +0.753] *(Llama-8B). Three of the five point* toward *a step detour and none of those three excludes* 0*; the largest of them (Llama-8B) also shows a censoring signature (late-solve fraction* 0.500 *vs.* 0.325*), whereas the 14B reversal does not (*0.391 *vs.* 0.434*). The honest summary is that the step channel is* uninformative *at this corpus and these sample sizes—not that it refutes the hypothesis. We therefore do* not *claim a measured step-detour penalty. One defense of the hypothesis was that our corpora contain no longer-route (latent) poison, so the contrast above is off-target. Appendix C narrows that defense: it* constructs *such a poison—bit-identical output, strictly longer prescribed route—so the mode is no longer merely hypothetical, and it reports that the agent does execute the longer route (*0.962 *of traces) while the executed-command trace shows a cost landing in operations packed per issued command—a channel e, which counts* rounds*, does not read. That probe does* not *measure the step cost: its round budget censors half its helpful and detour runs, so its marginal round contrast is confounded with solve status (corr* −0.84*) and it has no positive control validated for the round channel. The hypothesis therefore remains open and untested against its target mode; what has changed is that the construction now exists, so it* can *be tested.*

Proposition 3 (Poison's realizable outcome harm: displacement plus sub-floor)**.** *Relaxing A2 to allow displacement with probability* δ_f*, and A1 to allow a sub-floor deviation* $s_f \geq 0$ *(measured* 0 *at 3B/7B/Llama-8B—3B's deviation is* above *the floor and so contributes no sub-floor term—with small point estimates of* 0.012 *at 14B and* 0.040 *at 32B, the values we plug in below), the on-target poison* outcome *harm obeys*

$$\left|\mathrm{ATE}_{\mathrm{harm},o}(v \mid f)\right| \;\leq\; (h_f - b_f)\,\delta_f + s_f \;\leq\; (1-b_f)\,\delta_f + s_f,$$

i.e. the displacement term is the helpful lift *that poison crowds out, not a function of* b_f *alone, and the bound reduces to* s_f *as* $\delta_f \to 0$*. This is the entire realizable outcome harm in our data; both terms are small and neither is a base-rate gradient. On the* reward*, any separate step-channel term would require the step-detour of Hypothesis 1, which the data do* not *support—the measured step contrast is uninformative (reversed at 14B, null elsewhere, three of five rungs pointing the other way without significance)—so we claim no additional reward-channel harm, while noting that a null step channel is an absence of evidence here rather than evidence of absence.*

Proof sketch. Citing poison changes o only by replacing a helpful recipe (which would have lifted the floor toward h_f) with a non-reaching one (which returns to b_f); the lost mass per displacement event is the forgone helpful lift $h_f - b_f$, so the displacement harm is $-(h_f - b_f)\delta_f$, bounded in magnitude by $(1 - b_f)\delta_f$; the measured sub-floor deviation of A1's restatement enters additively as s_f. The reward channel adds no established term: the measured step contrast does not support Hypothesis 1 and is too uninformative to establish one. □

What restores identification (a different, recoverable estimand).

Proposition 4 (Execution verification escapes the floor and identifies recipe validity)**.** *Introduce the deterministic*

execution observable

$$o_{\text{exec}} \;=\; \mathbf{1}\left\{\begin{array}{l}\textit{the cited recipe, re-run in isolation,}\\ \textit{executes and yields a well-formed flag}\end{array}\right\}.$$

A benign-wrong recipe has $o_{\text{exec}} = 0$ by construction*; a helpful recipe has* $o_{\text{exec}} = 1$*. Because* o_{exec} *is a deterministic function of the recipe and does* not *marginalize over the agent's recovery path, it is outside A1's scope (A1 constrains only the run outcome* o*): the credited signal is the recipe's own success, which separates poison from helpful with probability* 1 *regardless of* b_f*. Conditioning credit on* o_{exec} *therefore yields a consistent, correctly-signed estimator for the target it measures even under A1–A2.* Caveat on estimand: o_{exec} *identifies* recipe validity *(a property of the lesson in isolation),* not *the citation-ATE on run reward that Prop. 2 declared non-identifiable from outcomes. These are different, complementary targets; the point is that recipe validity is recoverable and is the right instrument for flagging poison.*

Proof sketch. A1 held only because the run *outcome* marginalizes over the recovery path, equating cited poison with no recipe. o_{exec} isolates the recipe and reads its direct effect: 0 for a non-reaching poison recipe, 1 for a reaching helpful one, for any b_f. Hence $o_{\text{exec}} \not\perp \{v \text{ poison}\} \mid f$, the law-equality that drove Prop. 2 fails, and the contrast on o_{exec} has a nonzero estimand any consistent contrast recovers. *Empirical check (construction-guaranteed—the benchmark builds non-reaching poison recipes, so this is a deployable contrast on recipe validity, not a measured generalization):* the verifier flags poison on 100% and false-flags helpful on 0% of 35 benign families (and the same 100/0 separation replicates on the 41-family 14B malignant set). *Scope of* $o_{\text{exec}} = 0$*:* that a benign-wrong recipe fails to yield a well-formed flag is a property of *this* construction and of a domain admitting a per-output validity oracle (here the `picoCTF{}` predicate), not of benign-wrong recipes in general. On a separate author-built SQLite domain—distinct from the `campaign_sql_14b` domain of §9—which admits no such oracle, 27.8% of sibling-poison recipes are *accepted outright* by the same verifier, input-sensitivity included ($n = 320$; artifact listed in Appendix B). The 100/0 separation above is therefore a property of the crypto construction and its oracle. □

Reconcile with data. The model reproduces every salient pattern in §8—some as built-in consistency (its premises were measured on the same campaigns), and three as over-identifying checks that falsify a cruder "harm shrinks with the base rate" story. *(Helpful identified, larger at higher capability.)* Prop. 1 predicts a positive, detectable helpful effect; the observed on-target helpful solve ATEs are +0.247/+0.195/+0.526/+0.570/+0.227 across 3B/7B/14B/32B/Llama-8B (all $p < .001$, SNIPW CIs excluding 0), and the lift $h_f - b_f$ tends to grow with capability (non-monotonically—a 7B dip, a lower Llama-8B rung) as more capable solvers both raise h_f and execute recipes more reliably (cite/follow rates rising toward 1.0). *(Poison outcome non-identified.)* The on-target nulls at 3B/7B/Llama (+0.021, p=.68; −0.005, p=.82; −0.003, p=.75) restate A1's own empirical check on the same campaigns, so we count them as consistency, not independent confirmation; the marginal-inclusion nulls are a different estimand (−0.011, p=.45; +0.010, p=.42; −0.008, p=.49; all CIs include 0), but a *qualified* check: pooled over off-target retrievals the same estimator is non-significant on *helpful* at 3B (+0.009, p=0.255) and Llama-8B (+0.007, p=0.545), against +0.038/+0.060/+0.067 at 7B/14B/32B, so at two of those three rungs it cannot see a lesson we know is large on-target, and a poison null there is guaranteed. At 14B it *does* have demonstrated power on this estimand: a pre-registered ordinary-noise control that rides along in the same campaign (category-matched crypto lessons that are sampled but can never match a trace signature; $n = 4264$ runs) reads on-target +0.015, p=.65 as predicted, but *marginal-inclusion* −0.018, CI [−0.029, −0.008], p=.001 over 41 families. The marginal estimator therefore sees ordinary context clutter at this scale in the very cell where it reads poison null, which strengthens rather than weakens that null. We still do not treat the poison marginals as independent confirmation—and at 14B/32B the small negatives are not rankable per-item against the assumption-light leave-one-out gold standard (within-poison rank correlation ≈ 0, mean marginal CI including 0; §8)—it is the gold standard's inability to rank, not a randomization test, on which we rest the non-identifiability. *(Three facts a base-rate story cannot explain.)* (i) The harm does *not* track the family base rate at four of the five rungs (Spearman(b_f, poison ATE): 14B −0.033, p=0.843, n=39; 32B −0.178, p=0.260, n=42; 7B +0.068, p=0.708, n=33; Llama-8B +0.460, p=0.154, n=11); only the 14B and 32B cells are powered. The exception is 3B, where the gradient *is* significantly negative (−0.705, p=0.010, n=12)—and we note the Llama-8B cell running the other way rests on a comparable n=11; we report it rather than rest on the powered rungs alone. The powered-rung nulls are predicted, because by Lemma 1 the population outcome harm is 0 independent of b_f, and by Prop. 3 the only residue scales with the *helpful lift* and δ_f, not with b_f. (ii) The small significant *on-target* (family-restricted) negatives at 14B/32B (−0.057, p<.001, CI [−0.087, −0.022]; −0.064, p=.005, CI [−0.109, −0.018]) are quantitatively reconstructed by a *measured* two-component decomposition—displacement of the helpful lift plus the small sub-floor—using the directly-measured co-citation $\widehat{\delta}_f$ (not a back-fit): $-[(h_f - b_f)\widehat{\delta}_f + \text{subfloor}_f]$ gives $-[0.526{\cdot}0.062 + 0.012] = -0.045$ at 14B (within the CI of the observed −0.057) and $-[0.570{\cdot}0.050 + 0.040] = -0.069$ at 32B (vs. observed −0.064, almost exact). The measured displacement term alone covers ≈ 57% (14B) / ≈ 44% (32B) of the point estimate, the sub-floor the rest—so the residue is displacement-plus-sub-floor, *not* a base-rate gradient. These cells are the finite-δ_f regime of Prop. 3, *not* Lemma 1's exact-zero limit. (iii) Poison "solve | cited"

is at the floor at 7B/14B/Llama-8B, mildly *above* it at 3B, and mildly *below* it at 32B (-0.040, CI $[-0.062, -0.016]$): consistent with A1 as restated (exact at three rungs, $\leq$ 4pp sub-floor at 32B), the same small sub-floor harm as the M=1 rival of §8. *(A reward "2×" that is an identity, not a detour.)* The on-target poison *reward* ATE is 1.7–2.1× the outcome ATE (14B -0.119, 2.1×; 32B -0.106), but the *same* 2× holds for helpful lessons (2.12 vs. 2.09) and the direct conditional step contrast is uninformative at all five rungs (Hypothesis 1), so this is an R-scale identity, not a measured step-detour. Finally the recovery (Prop. 4) matches the 100%/0% verifier separation.

The randomization test, disclosed and demoted. An earlier draft leaned on an "exact" conditional-randomization test (CRT) keyed on o as assumption-light corroboration. We retract that reliance. The independent-Bernoulli CRT is *anti-conservative* for top-m sampling-without-replacement (measured Type-I 0.11 at 14B / 0.17 at 32B against nominal 0.05), is *underpowered* (power 0.38–0.44 for a -0.057 effect), and is *blind* on our positive control (it returns frac-sig 0.000 even on genuinely destructive destroy/exhaust harm). It is therefore non-diagnostic here and we use it for nothing: helpful identification rests on the SNIPW magnitude, and poison non-identifiability on the leave-one-out gold standard's inability to rank.

Scope and honesty. This is an identification/power result *under* A1 (measured; exact at three rungs, above-floor at 3B, $\leq$ 4pp-approximate at 32B) and A2 (scope-delimiting); H1 is a *hypothesis* the step data neither support nor refute, not a premise we use. It is specific to benign-wrong (non-reaching) lessons under near single-item retrieval. The content is its consequence for the entire class of *consistent outcome-based estimators*, not the trivial zeroing of one contrast. It does *not* claim harm is always zero (Prop. 3 gives the displacement channel, and there is a measured $\leq$ 4pp sub-floor at 32B and at M=1), and does *not* claim outcomes can never detect poison in richer regimes: actively destructive lessons that push outcomes *below* b_f violate A1 and *are* detected (the positive control). It says: in this measured regime, the benign-wrong lesson's per-item harm is not identifiable from the binary verifiable outcome, so outcome-based credit is the wrong instrument, and a direct execution observable (recipe validity) is the right one. Generality is bounded: the capability ladder is Qwen 3B–32B; the only non-Qwen rung not harness-floored is Llama-3.1-8B (Gemma-27B, Mistral-7B/24B, and InternLM-7B probes floored at $n \leq 3$), so "across architecture" here means one architecture.

Falsifier: Exhibit an on-target benign-wrong (recovered-from) lesson whose cited solve rate is detectably *below* b_f by materially more than the $\leq$ 4pp the 32B cell already shows (a large sub-floor harm would refute A1), or an outcome-based estimator that *ranks* the per-item poison effect in agreement with the leave-one-out gold standard (which would refute the non-identifiability). A small reward ATE with a null/at-floor outcome, as at 14B/32B, is an R-scale identity, not a falsifier. For the reward channel specifically, exhibit a longer-route (latent) poison whose route cost is measurable in the reward's accounting unit. Appendix C constructs such a poison and finds the cost visible in the executed-command trace but not in rounds; its design cannot resolve whether a round cost exists, so this falsifier is open and now has an instrument to be attempted on.

9 Discussion and Limitations

This work pairs a 4B *pilot* (which illustrates the boundary) with a lab-scale ladder (which carries the headline), and several limitations bound the claims. **The strongest rival reading: a significant *on-target* sub-floor harm.** Our headline is the *per-item* non-identifiability of §8.1: the gold standard cannot *rank* poison lessons. The deployed *aggregate* quantity—the *marginal-inclusion* ATE, the causal effect of *including* a poison lesson in the retrieved set—is additionally null at every retrieval breadth (M=1 -0.021, p=.25; M=3 $+0.004$, p=.79; M=6 -0.002, p=.86). The most honest rival is the *on-target* contrast, which restricts the run set to challenges of the poison lesson's own family: it is a small sub-floor harm whose point estimate is roughly constant across breadth (M=1 -0.0745, $p < .001$, n=9 lessons, CI $[-0.125, -0.033]$; M=3 -0.057, p<.001; M=6 -0.054, p=.11, CI $[-0.116, +0.011]$), significant at the powered M=1/M=3 rungs but *not* at M=6 (where its CI includes 0). So within crypto, *when a poison lesson is executed*, it does cost a little solve probability—a real harm, conditional on family, not on citation. Two honest distinctions keep this from touching the headline. First, the *marginal* contrast—what a deployed memory actually controls—pools over the off-target retrievals it also makes and is null at every M, whereas the on-target contrast conditions on the challenge being one the lesson targets, which deployment does not guarantee. Both are causal effects of the same logged inclusion draw; they differ in conditioning set, not in selection (family is pre-treatment). Second, even where the on-target contrast is significant (M=1 and M=3), the gold standard cannot *rank* which poison lesson is worse (within-poison rank correlation ≈ 0): the aggregate conditional harm is real but not attributable per item. The per-item non-identifiability the paper rests on therefore holds at every M. **Scale and statistics.** The *pilot* is one 4B model, ten families, tens of tasks (the 80 rows are 40 challenges × two near-duplicate seeds, identical B0 outcome on 37/40, so we report at the challenge level and treat n as $\approx$40). The lab-scale ladder that carries the headline (§6) is larger by orders of magnitude: five capability rungs (Qwen2.5 3B–32B plus Llama-3.1-8B), 2.2k–15.6k logged runs per rung over 11–43 families ($\sim$15.6k/15.5k at the powered 14B/32B rungs). **A minimal instrument,**

not wild CTF. Each task is a single fixed-transform decode solvable in one to four shell rounds—closer to a sanity-world than to Cybench/NYU-CTF-scale exploitation [23, 37]. Its value is the *known* ground-truth lesson causality, not realism; an external-validity probe on live picoCTF (blocked by bot-protection) is in Appendix A. **One domain, and a second where the null fails.** All our non-identifiability claims are scoped to a *single* hand-built crypto-decode microbenchmark—the regime where a benign-wrong lesson yields self-evidently-wrong, recoverable output. In a second, SQLite filter-path domain the deployed-M null does *not* replicate: poison lowers solve on *both* estimands (on-target -0.25, p=.003; marginal-inclusion -0.065, p=.011).[9] There the helpful lesson coincides with the exact answer-key predicate (base ≈0, helpful solve ≈1.0), even the poison harm is *displacement* of the correct query (co-presence with the helpful lesson does not lower solve)—context-dependence, not an identifiable per-item harm. Read through §8.1, this break is consistent with Prop. 3's displacement channel at an extreme helpful lift ($h_f - b_f \approx 1$ makes even modest displacement detectable, with the floor premise itself intact at $b_f \approx 0$): what fails to transfer is A2's small-displacement scope, not the identification logic. And the signal is regime-fragile (a larger same-domain re-run floored to an exact null). **And a third that degenerates.** A struct/tag-extraction domain self-verifies through the `picoCTF{` prefix and is brute-forceable. We therefore do *not* claim the boundary generalizes across domains; whether it holds beyond self-evidently-recoverable decode tasks is open, and the failed SQL generalization is the most important caveat on external validity. **The positive control over-samples.** The malignant campaign over-samples its planted lessons ∼2.5×, depressing the base rate so that same-campaign benign poison reads spuriously positive (+0.033); the benign-poison null is therefore the *main*-campaign result (§8), and the destroy/exhaust detection survives only because it is floor-violating. **Solve rate does not separate the credit signals, and we do not close the loop to where it would.** As §8 explains, a capable solver follows the true lesson whenever it is co-retrieved, so the bandit conditions sit at solve-parity. We are deliberately careful about what *does* separate them: the signals assign poison different learned *values*, but we have shown that difference is a confounded sign, not an identified per-item effect, so we do *not* claim a demonstrated credit-*quality* difference on benign poison. We do *not* run write-back end-to-end under *live* retrieval. A pre-registered forced-exposure transfer probe does test the promotion *decision* downstream—arms differing only in whether each credit signal would have promoted the poison, evaluated on a split where the family's true lesson is withheld—and it finds no causal-credit advantage: Δsolve (causal − outcome memory) is −0.32pp, CI $[-2.08, +1.44]$, n=624 at 14B (item-level verdict *null*; clustered over 39 families the interval widens to $[-3.04, +1.60]$, p=.94, verdict *inconclusive*), and −1.49pp, CI $[-2.83, -0.30]$, McNemar p=.041, n=672 at 32B (item-level *inconclusive*; clustered over 42 families, $[-4.46, +0.45]$, p=.27). Once clustered the probe is *inconclusive at both rungs*. Both memory arms are also indistinguishable from a matched-budget *shuffled* memory. Retrieval is forced there, so it brackets a worst case rather than measuring deployed behaviour, and the cost at scale under live retrieval remains unmeasured; and because our promotion-rate proxy rests on that same confounded sign (§8), it is an illustration of the failure mode, not a measured suppression. Two pilot levers also remain unvalidated against their target mode. Gating is measured only by its provisional/trusted split. The step-count reward targets longer-route (latent) poison, which neither corpus instantiates; Appendix C constructs that mode and shows the agent executes the longer route, and that a cost appears in the *density* of its executed commands—a channel e does not count. Whether it *also* costs rounds is unresolvable there: that probe's round budget censors half its helpful and detour runs, so it does not measure the step cost and we do not claim it as a test of the lever. **Active re-rolls are expensive; the observational signal is not—but its assumptions are real.** The active leave-one-out probe costs extra solves and stays a diagnostic. The observational estimator needs no extra inference and *is* folded into an online bandit here, recomputing its values from logged exploration each epoch; its caveats (interference, positivity) are detailed below. Estimates are over n=4 lessons per label, so we report direction and sign, not precise effect sizes. **Coarse context.** The bandit context is three category nodes (crypto/forensics/misc), so within-category lesson selection is unpersonalized; the spec's cluster context may separate methods better and is untested. **Familiarity, not just contamination.** Random flags remove flag look-up, but the *methods* (Base64, ROT13, `strings`) are ubiquitous in pretraining; this is precisely why five families sit at ceiling and is why capability-dependent modes (how often a model recovers from executed poison) may differ at 27B and cannot be inferred from a 4B pilot. **Scope of claim.** We characterize false-lesson *exposure within the verifier-detectable subset*; model-internal shortcuts that pass the checker without the stated mechanism are out of scope and bounded by the random-flag design.

Limitations of the causal-credit estimator (§5). The observational variant is the contribution most in need of caveats, and we state them plainly. **Interference (SUTVA).** The run reward depends on the whole sampled M-set, so the observational ATE is a marginal effect averaged over co-sampled lessons, not an isolated treatment effect; coarse context can correlate the M-sets within a cell. Our mitigation is structural: the active re-roll (§5.1) holds the other sampled lessons fixed and is therefore interference-robust, so we use it to *validate* the cheap estimator rather than treating the observational number as assumption-free.

[9] `campaign_sql_14b/merged/ontarget_analysis.json`: on-target poison −0.2513, p=.0027, n=8; marginal-inclusion −0.0646, p=.011, n=16 (both exclude 0).

Non-additivity. When lessons interact (e.g. complementary pairs that solve only together), the exact ATE is attenuated; we verify on an engineered non-additive world that the *helpful* ranking the estimator does recover survives the interaction. We do *not* claim a reliable true-above-*poison* per-item ranking: the gold standard shows the per-item poison effect is unrankable in this regime (§8), so that ordering is not a property we assert. **Positivity/overlap.** As the bandit sharpens, inclusion propensities approach 0 or 1 and weights explode; we use self-normalized IPW, reject cells whose effective sample size falls below a fraction of their count, clip propensities to $[0.05, 0.95]$, require ≥ 5 observations per arm, and partially pool sparse cells toward the global estimate. **Propensity circularity.** The propensities are estimated from the same scores that drove sampling; a fully unconfounded estimate would log an independent exploration stream, which we do not. These are real bounds on a method whose appeal is its zero marginal cost. What it recovers for free—the large value of *helpful* lessons, which outcome credit cannot separate from poison at *any* amount of the same logged data—is genuine; what it does *not* recover—the per-item harm of a benign-false lesson—is the identification boundary of §8.1: at 3B, 7B and Llama-8B, where the measured sub-floor is zero (3B's deviation is *above* the floor), the outcome estimand is not distinguishable from zero and no sample size closes it; at 14B and 32B, where the decomposition plugs in a small *point* sub-floor (0.012, CI $[-0.029, +0.007]$; 0.040, CI $[-0.062, -0.016]$), and at $M{=}1$, where the on-target harm is -0.0745 but no A1 contrast was run, the aggregate label-level harm *is* detected at large n and only the *per-item ranking* remains beyond any feasible re-roll budget (§8).

10 Conclusion

A verifiable reward tells you *that* a run succeeded, not *which* retrieved lesson made it succeed—so an executed false lesson the agent recovers from is credited exactly like a useful one, and an outcome-credit memory cannot tell them apart. The same setting hands you a free randomized experiment—retrieval is stochastic with a logged propensity and the reward is re-checkable—and our contribution is to map exactly how far it reaches. For *helpful* lessons it reaches all the way: inverse-propensity weighting over the existing logs recovers their large marginal value at zero extra inference, in agreement with an active leave-one-out gold standard and broadly (non-monotonically) growing with capability. For the per-item harm of a *benign-false* lesson the agent recovers from, it does not reach at all: neither the cheap estimate nor the gold standard can rank the effect, and we argue why—under an empirically-measured (at 32B only approximate) base-rate floor, executing a benign-wrong recipe is, for the binary outcome, nearly indistinguishable from executing none, so no consistent outcome-based estimator can isolate it. A positive control draws the boundary's other edge: actively-destructive lessons that breach the floor *are* detected, a confidently-wrong answer is not (though at a follow rate of 0.024 that arm is untested rather than shown to be invisible), and recipe validity is recoverable by a deterministic execution-verifier. The practical upshot is a caution, not a tool: the mildly-negative sign an observational estimator assigns benign poison is confounded, so a write-back gate built on it is not trustworthy. We also draw a sharper boundary than expected on *where* false lessons matter—a capable solver follows the true lesson when it is co-retrieved, so they do not lower immediate solve rate at deployed retrieval (a small conditional-on-execution harm remains; §9); any latent cost lives in the persistent memory and would surface through write-back, whose downstream cross-task solve-rate cost we measure only under forced exposure (§9), not at scale under live retrieval. What the verifiable-reward setting gives for free is the ability to credit large useful lessons and to flag actively-destructive (floor-breaching) ones—though not a confidently-wrong answer, which our agent followed on 11 of 453 sampled runs and which we therefore leave untested at deployed exposure—under *forced* exposure both a constructed leave-one-out (-0.43) and the deterministic execution-verifier catch it; what it does *not* give—and what we argue it cannot give from outcomes alone—is the per-item harm of a benign-false lesson. The open problems are to quantify the downstream cost of propagation at lab scale (the 27B solver, the full CTF sets) and to find any oracle-free signal that ranks benign-false harm; the released harness changes only its backend.

Author contributions, AI use, and acknowledgments

Author contributions. J.H. designed and implemented the system, ran the experiments, and prepared the manuscript. X.L. supervised the work. S.J.C. contributed to the research; the authors will state the specific division before publication. Appendix C was carried out by J.H. alone, without the involvement of the other authors, as a graduate course project built on the shared system of §6; J.H. is responsible for its inclusion.

Use of AI assistance. Claude (Anthropic), via the Claude Code CLI harness documented in the released repository, was used substantially in producing this work between 2026-05 and 2026-08. We state the extent plainly rather than as a formality. It was used to write and refactor code across the research system—the knowledge-graph store, the retrieval and sampling pipeline, the bandit and reward implementations, the evaluation and off-policy-estimation harnesses, and the analysis and figure scripts—and to draft and revise the text of this manuscript, including automated critique-and-revision passes over the draft. Fourteen of the repository's 173 commits carry

the assistant as author; that attribution *understates* its involvement, because later work—including manuscript rewriting and the critique passes—was committed under the first author's name. No figure in this paper is AI-generated; every figure is rendered by the released scripts from committed run artifacts.

The division of authority is the part that matters. The research question, the integrity constraints that governed the campaigns, the acceptance criteria, and the final adjudication of what this paper asserts are the authors'. Reported numbers are either macro-expanded from emitter output or transcribed from a named committed artifact, and the authors verified them against those artifacts. Two qualifications belong here rather than in a footnote. First, several experimental-design parameters for Appendix C—the solver model, the output-token limit, the round-budget regime, and the family-selection rule—were chosen by the assistant during unattended operation. Each is logged with its reason, and with when it was fixed relative to the contrast it could affect, as D-01–D-04 of `appendix_c/APPENDIX_C_DECISIONS.md`. No analysis reported in that appendix was chosen after seeing a result it could affect. Second, AI assistance is not an author, holds no responsibility for the content, and the authors take full responsibility for any error that survives.

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A External-validity probe: live picoCTF

For external validity we attempted the same pipeline against live `play.picoctf.org` with valid account credentials, a real (non-headless-UA) browser, and the project's scraper. The site is served behind a commercial bot-mitigation service: the login redirected to an interstitial that "verifies you are not a bot," and the practice listing was gated the same way. We made a single legitimate browser-plus-credential attempt and did *not* deploy anti-automation evasion (stealth drivers, CAPTCHA-solving) to bypass the access control. This is consistent with our design argument: live public CTF is not only *contaminated* (the model may have seen the flags) and *label-free* (no ground-truth lesson causality), but also *operationally gated* against automation. The contamination-free, ground-truth-labeled corpus is therefore the appropriate primary instrument; lab-scale evaluation on sanctioned benchmarks [23, 30, 37] with the 27B solver is future work.

B Reproducibility

The code, the benchmark generator, and every artifact referenced below are released at `https://github.com/jawh3/solver-graph`. The pilot numbers (§7) are emitted by `experiments/analyze.py` into `paper/numbers.tex` from `results/pilot/`. The causal-credit study (§8) is run by `experiments/run_campaign.py` (conditions B0/B2/outcome/observational over the recalibrated corpus) into `results/campaign_v2/`; `experiments/analyze_campaign.py` emits every §8 number into `paper/numbers_solution.tex` (held-out solve rates with paired bootstrap and exact McNemar, the outcome bandit's learned Q by label, the observational IPW ATE re-scored from the same logs, and the promotion-rate proxy), and `experiments/fig_solution.py` regenerates the figures. The recalibrated corpus—outer transport encodings wrapping invisible inner transforms, with fresh random flags and ground-truth helpful/poison/distractor labels—is generated deterministically by `benchmark/hard_ctf.py`; the observational estimator and its hardening (self-normalized IPW, effective-sample-size rejection, propensity clipping, partial pooling) are in `ope/causal_value.py`. The model is served by `local_llm/server.py` (4-bit, OpenAI-compatible). *Note:* the `campaign_v2` pilot was run on an earlier benchmark revision whose poison-contract, oracle-labeling, and grid-duplication bugs were fixed in commit `63db02d` and was not re-run, so its 4B magnitudes are illustrative; the load-bearing numbers are the lab-scale ladder below. The *lab-scale* ladder (§8, §8.1) lives under `results/lab/campaign_{3b,llama8b}_M3`, `campaign_7b35_M3_rep`, `campaign_14b_M3_powered`, `campaign_32b_M3_tp2` (the 32B powered rerun), the malignant control `campaign_14b_malig(_expand)`, and the M-sweep `campaign_14b_M{1,6}`; the headline C1 statistic is emitted by `experiments/within_poison_validation.py` (into `active_loo_within_label.json`), the interference/DiD by `experiments/crypto_interference_analysis.py`, and the per-rung macros by `experiments/emit_paper_numbers.py`. The harness changes only its backend between the single-GPU pilot and the lab-scale ladder. The Appendix C probe is separate from both and lives under `appendix_c/`: the construction and verifier check in `appendix_c/results/g1_verifier_acceptance_crypto_depth1.js` (and the SQL cross-check of Prop. 4 in `g1_verifier_acceptance_sql.json`), the paired step probe's per-run rows in `appendix_c/results/g2_step_probe/`, the command-recording re-run in `appendix_c/results/g2_mechanism/` (per-run rows; the aggregate `g2_mechanism.json` reproduces from them exactly), and the family selection in `appendix_c/results/inband_selection_crypto.json`; each carries a companion `.provenance.json` recording run id, seeds, model and timestamp, and every Appendix C macro is re-derived from them by `emit_appendix_numbers.py`. Parameter choices made during that probe are logged with their reasons in `appendix_c/APPENDIX_C_DECISIONS.md`. *Pre-registered arms not reported above.* Two ran and are recorded here for completeness. A name-leak A/B at 3B (`campaign_3b_nameab_{leak,fixed}`, 2×2016 runs) tested whether leaked lesson names induce a spurious on-target poison negative; the leaky cell is *null* (+0.013, CI $[-0.074, +0.114]$, p=.82), so the pre-registered falsifier fires and the leakage artifact is capability-dependent—the demonstrated instance remains the 7B pair, and this manuscript makes no name-leak claim. The ordinary-noise distractor control (T2-3, n=4264) is reported in §8.1.

C A constructed longer-route poison, and what the step channel sees of it

§4 proposes the step-count reward $R = o(1+\lambda e)$ as the lever aimed at *latent poison*—a lesson whose premise is false but whose route still reaches the flag. Neither the pilot corpus nor the lab-scale corpus instantiates that mode: their poison is a *non-reaching* wrong tool, so Hypothesis 1 rested on the defense that the measured step contrast was off-target. This appendix removes that defense by *constructing* the missing mode, and reports what a first probe of the step channel against it does and does not establish.

Two of its three results are clean, and one is not. The construction and the execution-verifier's blindness to it are structural. The mechanism—the agent executes the longer route and absorbs it inside a round—is a direct observation of the command trace, though only its execution rate and its command *density*, not the raw command count, survive conditioning on outcome. The step *contrast*, by contrast, is heavily censored by the round budget and we report it as largely uninformative rather than as a bound; saying so is the point of §C.2.

Scope. One model (Qwen/Qwen2.5-7B-Instruct), one crypto domain, 13 families, 52 paired challenges. It uses the same model as the 7B rung of §6 but a different design—forced single-lesson context, a flat 8-round budget, and 13 selected depth-1 families—so it is *not* a rung of that ladder. It is a marginal and conditional step contrast, not a per-item ranking: the leave-one-out gold standard that carries the §8 headline was *not* run against this construction, so nothing here speaks to per-item identifiability. Reported at appendix strength; it enters no headline claim.

C.1 Construction

The payload is an *involution pad*. Where the correct recipe for a family is a transform chain T, the poison recipe (the `detour` arm) prescribes $\pi \circ \pi \circ T$ for a self-inverse π on the alphabet (we use rot13; the same construction holds for $\texttt{xor}_k$ and `rev`). Because $\pi \circ \pi$ is the identity, the recovered flag is *bit-identical* to the correct method's while the prescribed route is strictly longer. The lesson is false in the paper's own sense (§3): it asserts a causal premise—that the pad is required—which is wrong for the challenge at hand.

Over 52 depth-1 crypto families × 8 instances ($n = 416$), output identity holds on 416/416 and the route is strictly longer on 416/416, by exactly two extra steps on every instance.[10] A second arm (`search`) prescribes an exhaustive trial over a candidate set containing the truth—longer in expectation but not output-identical. A third (`sibling_poison`) names a plausible wrong transform that does not reach the flag; it is the corpus poison type of §6, included here as a reference.

[10] The `--depth 1` restriction is load-bearing: the construction is defined only where a family's chain admits the pad at depth one.

Table 3: Per-condition marginals, $n = 52$ each, 8-round budget, Qwen/Qwen2.5-7B-Instruct. "Cap" is the fraction of runs that exhaust the budget. Note that solved runs finish fast and unsolved runs sit at the cap, so the round mean is largely a restatement of the solve rate.

Condition	Solve	Rounds	Cmds	Cap
None (no lesson)	0.019	7.79	7.21	0.904
Helpful	0.365	5.38	4.67	0.577
Detour	0.404	5.13	5.42	0.481
Search	0.077	7.46	8.54	0.904
Sibling poison	0.019	7.69	6.21	0.904

The execution-verifier is blind to it, by construction. On this crypto corpus the verifier of Prop. 4 accepts the detour arm at 1.000, the same rate as a genuinely helpful lesson, while flagging `sibling_poison` at 1.000, `destroy` at 1.000, `exhaust` at 1.000, and the confidently-wrong `wrongflag` arm at 1.000 (the last is well-formed at 1.000 and fails only input-sensitivity).

This acceptance is *near-tautological*, and that is the finding rather than a defect. The verifier's criterion is well-formedness plus input-sensitivity; a recipe that recovers the true flag satisfies both however long its route. Prop. 4's reach is bounded exactly here: it identifies recipe *validity*, and a longer-route recipe is valid. The measurement's value is that it makes that boundary a demonstrated property of the deployed verifier rather than an argument about it.

C.2 The step probe, and why its marginal contrast is not usable

We probe with a paired, forced-context design: each of 52 challenges (13 families × four instances) is run once per condition on a *common seed* (0 seed mismatches across all 52 challenges), under a flat 8-round budget. Families are the 13 of 32 selected in-band by a calibration probe that never saw a latent-poison condition. Table 3 gives the per-condition marginals; Table 4 gives the paired contrasts.

The marginal round contrast is a restatement of the solve rate. The budget censors heavily: 30/52 helpful and 25/52 detour runs exhaust it, and 20 pairs (0.385) are capped in *both* arms and therefore contribute $\Delta = 0$ by construction. Meanwhile solved runs finish in 1.37 (helpful) and 1.62 (detour) rounds while unsolved runs sit near the cap. The round variable is thus close to a two-valued function of the outcome, and the paired contrasts confirm it: $\text{corr}(\Delta\text{rounds}, \Delta\text{solved}) = -0.84$ for the detour arm and -0.95 for search. The marginal -0.250 rounds is therefore *not* a measurement of route cost, and we do

Table 4: Paired contrasts against HELPFUL on common seeds. Intervals are 95 % *family-clustered* paired bootstraps over 13 clusters—a small cluster count, so they are anti-conservative and we do not treat any endpoint as a hard exclusion. Clustering widens five of the six marginal intervals relative to an i.i.d. resample. The *marginal* row is reported for completeness only; see the text for why the *both-solved* row is the interpretable estimand.

	Detour	Search
Marginal ($n = 52$ pairs) — confounded		
Δ rounds	-0.250	$+2.077$
95 % CI	$[-1.212, +0.654]$	$[+0.827, +3.538]$
Δ solve	$+0.038$	-0.288
95 % CI	$[-0.135, +0.231]$	$[-0.500, -0.096]$
disc., p	5/7, p=0.774	15/0, p<0.001
corr($\Delta r, \Delta o$)	-0.84	-0.95
Both-solved — uncensored		
n (families)	14 (4)	4 (2)
Δ rounds	$+0.071$	$+0.500$
95 % CI	$[+0.000, +0.200]$	$[+0.000, +0.667]$

not read it as one. This is the same censoring failure the companion project's design guidance warned against, and the paper's own corpus step contrast (§8.1) is reported conditionally on solved runs for exactly this reason.

What the design can and cannot resolve. On the uncensored both-solved estimand the detour arm costs $+0.071$ rounds ($[+0.000, +0.200]$), a very small positive cost—but from 14 pairs spanning only 4 families, which is too thin to carry weight. A censoring sensitivity check shows the scale of the problem directly: shifting the helpful arm's own round profile by a constant and re-censoring at the cap, a *true* cost of one round per run would be observable as only $+0.423$ rounds, and two rounds per run as $+0.846$. A one-round per-run cost is well inside anything this design could distinguish from zero. We therefore make no claim to have bounded the step cost of the payload.

There is no validated positive control for the round channel. The `search` arm was intended as one, and on the marginal contrast it looks like one ($+2.077$ rounds). It is not: it is censored at 0.904, its round excess tracks its solve failure (-0.95), and on outcome-concordant pairs it falls to $+0.270$ $[+0.053, +0.550]$, against the detour arm's own concordant -0.150 $[-0.475, +0.136]$. That the two *do* separate on that estimand is not evidence the channel works: 30 of search's 37 concordant pairs are capped in *both* arms and contribute $\Delta = 0$ by construction, so the concordant estimand is diluted toward zero rather than uncensored. We treat neither arm's round contrast as validating the channel. What the search arm demonstrates is that this instrument is sensitive to *outcomes*; it does not establish that the instrument is sensitive to *route cost*. Note also that an arm which costs solves (-0.288, p<0.001) is by that fact already visible to the cheaper outcome channel of §8, so it could not have served as the interesting case regardless.

C.3 Mechanism: the route is executed, and absorbed

The one thing the probe establishes directly requires no round arithmetic. On a separate command-recording re-run over 26 of the 52 challenges at fresh seeds, the pad appears in the agent's *executed shell commands* on 0.962 of detour traces and 0.000 of helpful traces—and among *solved* runs alone, where no censoring can act, the separation is complete: 1.000 of 11 solved detour runs execute the pad against 0.000 of 12 solved helpful runs. The agent does not ignore the lesson; it runs the longer route.

The cost of doing so appears in the trace as *denser* commands, not more of them. Paired per-run shell operations packed into each issued command rise from 2.66 to 5.44, a difference of $+2.777$ (family-clustered CI $[+1.787, +3.834]$; 21 of 22 pairs positive, clustered over 12 families; 4 of the 26 re-run challenges drop out because one arm issued no shell command). The difference survives conditioning on outcome—restricted to both-solved pairs it is $+2.286$ $[+2.000, +2.750]$—but that restriction leaves only 7 pairs in 4 families, so no endpoint of that interval is a hard exclusion, and we hold its *width* to the same standard as the thin both-solved round estimand above. Its *status* is different, though, and that difference is the point: unlike the round channel, this one is not a restatement of solve status. On the same pairs, corr(Δops, Δsolved) $= +0.10$ against corr(Δrounds, Δsolved) $= -0.86$. What the restriction does show is that conditioning *shrinks* rather than inflates the estimate (the remaining 15 pairs give $+3.006$), and that its magnitude sits close to the two operations the pad prescribes—so conditional on the pad being executed it is close to arithmetic. The informative quantity here is the execution rate, not the size of the density shift.

The issued-command *count* does *not* separate the arms and we do not claim it does. Its paired difference on this re-run is $+0.409$ with a clustered interval spanning zero ($[-1.609, +2.526]$) and a median of -1.0; on the main probe's both-solved estimand the command contrast runs the *other* way (-0.786, $[-1.000, -0.400]$). Any marginal command difference here is produced by unsolved runs—the same confound as §C.2, and we flag it rather than spend it.

So one component of the execution trace—operation density within a command—plainly registers the longer route, while the reward's accounting unit does not: e counts *rounds*. That is the appendix's substantive observation, and it is an observation about the reward's *units*, not an estimate of an effect size.

C.4 Reading, and what it does to Hypothesis 1

The natural conjecture the construction suggests is that the relevant property is not route *length* but route *compressibility*: a longer route that collapses into the agent's existing accounting unit leaves the step channel flat, whatever λ is set to. We label this a *conjecture* and not a result. The mechanism data (§C.3) is consistent with it and is the reason to take it seriously; the step contrast (§C.2) neither confirms nor refutes it, because this design cannot resolve a per-run cost of the size at issue.

What the appendix does settle is narrower and still worth having. A longer-route poison *exists* and is now constructible on this corpus, so Hypothesis 1 is no longer defended by the absence of its target mode; and the deployed execution-verifier is blind to that mode by construction, which bounds Prop. 4 exactly where §C.1 says. Testing the step lever properly needs a design this one is not: an uncensored budget (or a round cap well above the correct method's run length), enough both-solved pairs across enough families to estimate a sub-round effect, and a positive control that costs rounds *without* costing solves. We state that as the work the construction now makes possible.

C.5 Limitations of this appendix

Censoring. The dominant limitation, treated in §C.2: at a flat 8-round budget 55 of the 104 helpful and detour runs are censored (196/260 across all five conditions), the marginal round contrast is confounded with solve status, and the uncensored estimand rests on 14 pairs from 4 families. **No round-channel control.** As above, the intended positive control validates outcome sensitivity only. **One model, one domain.** Qwen/Qwen2.5-7B-Instruct only; absorption is plausibly capability-dependent, since a solver that pipelines more aggressively hides more, and nothing here is measured at 3B, 14B, 32B, or on Llama-3.1-8B. **Family selection.** The 13 families were selected on a `none`-versus-`helpful` calibration probe at $n = 2$ per condition, thresholded on the *reference* arm, which the main probe then re-measured much lower (0.365); 4 of the 13 kept families re-measured at the floor (16 challenges, contributing only capped rounds). The resulting bias is toward a *smaller* helpful–detour gap—that is, it works *in favor of* the null this appendix reports, not against it. Part of that null may be a selection artifact and an unselected replication is required. **Estimand.** Marginal and conditional step contrasts, *not* the within-poison rank correlation against a leave-one-out gold standard in which the C1 headline of §8 is stated. No per-item ranking of latent poison was attempted, and none should be inferred. **Provenance.** The construction, probe, and analysis are the work of J.H. alone as stated in the author-contributions note preceding the bibliography; artifacts, seeds, and the re-derivation script are listed in Appendix B.